Measure Theory and Fine Properties of Functions

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1 August 26, 2024

Pop quiz: what's a measure? If you said "a function on sets," you're not wrong but you're not really right either. The whole game in this chapter is that you *can't* define a measure on all subsets of $\mathbb{R}^n$ and keep the properties you want. So you start with something weaker — an outer measure — and then cut down to the sets that behave.

Today we set up outer measures and Carathéodory's criterion.

1.1 Outer measures

The idea is: outer measures are cheap. You can define them on *every* subset. The cost is you only get subadditivity, not full additivity.

Definition 1.1 (Outer measure)**.** An **outer measure** on a set X is a function $\mu^*\colon \mathcal{P}(X) \to [0,\infty]$ satisfying

1. $\mu^*(\varnothing) = 0$,
2. (Monotonicity) $A \subseteq B \implies \mu^*(A) \le \mu^*(B)$,
3. (Countable subadditivity) $\mu^*\big(\bigcup_{k=1}^\infty A_k\big) \le \sum_{k=1}^\infty \mu^*(A_k)$.

That's it. No σ-algebra needed yet — we're measuring everything, just badly. The subadditivity is the price you pay for working with all of $\mathcal{P}(X)$.

1.2 Carathéodory's criterion

Here's where it gets interesting. We want to identify the "well-behaved" sets — the ones where the outer measure actually splits additively.

Definition 1.2 (μ^*-measurability)**.** A set $A \subseteq X$ is μ^***-measurable** if for every $E \subseteq X$,

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A).$$

This definition confused me for a while. Why test against *every* set E? Brian explained it like this: you already get $\le$ from subadditivity for free. So the real content is $\ge$. What you're asking is that A acts as a "clean cut" — when you slice any test set E along A, no mass leaks out. That's the whole point.

Theorem 1.3 (Carathéodory)**.** *Let μ^* be an outer measure on X. Then the collection*

$$\mathcal{M} = \{A \subseteq X : A \text{ is } \mu^*\text{-measurable}\}$$

is a σ-algebra, and μ^* restricted to $\mathcal{M}$ is a (complete) measure.

Proof sketch. Complements are free — the criterion is symmetric in A and A^c. For finite unions: take $A_1, A_2 \in \mathcal{M}$ and test against E. Split first by A_1, then split $E \cap A_1^c$ by A_2. You get three pieces that reassemble by subadditivity.

Now the hard part. Countable unions. Let $A = \bigcup_k A_k$ with the A_k disjoint (WLOG — replace A_k with $A_k \setminus \bigcup_{j<k} A_j$, which stays in $\mathcal{M}$ by the finite case). Set $B_n = \bigcup_{k=1}^n A_k$. Then for any E,

$$\mu^*(E \cap B_n) = \sum_{k=1}^n \mu^*(E \cap A_k)$$

by induction. So

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \setminus B_n) \geq \sum_{k=1}^{n} \mu^*(E \cap A_k) + \mu^*(E \setminus A),$$

since $A \supseteq B_n$. Let $n \to \infty$ and do the usual epsilon management. Completeness is free: any μ^*-null set passes the criterion trivially. □

Remark 1.4. That completeness comes for free is a big deal. If you build Lebesgue measure via Carathéodory, it's automatically complete — no need for the separate completion step you'd do starting from Borel sets.

1.3 σ-algebras and Borel sets

Definition 1.5. A σ-algebra on X is a collection $\mathcal{A} \subseteq \mathcal{P}(X)$ containing $\varnothing$, closed under complements, and closed under countable unions.

Nothing to see here. Closure under countable intersections follows from De Morgan.

Definition 1.6 (Borel σ-algebra)**.** For a topological space X, the **Borel σ-algebra** $\mathcal{B}(X)$ is the σ-algebra generated by the open sets.

In $\mathbb{R}^n$ you can equivalently generate $\mathcal{B}(\mathbb{R}^n)$ by open balls, closed sets, compact sets, or half-spaces $\{x : x_i < a\}$. Evans doesn't dwell on this. Every real analysis book covers it.

1.4 Radon measures

These regularity conditions are everywhere in later chapters, so I'm trying to internalize the hierarchy now rather than looking it up every time.

Definition 1.7. A **Borel measure** on $\mathbb{R}^n$ is a measure defined on $\mathcal{B}(\mathbb{R}^n)$.

Definition 1.8 (Borel regular)**.** A Borel measure μ is **Borel regular** if for every $A \subseteq \mathbb{R}^n$, there exists a Borel set $B \supseteq A$ with $\mu(B) = \mu^*(A)$.

Definition 1.9 (Radon measure)**.** A Borel regular measure μ on $\mathbb{R}^n$ is a **Radon measure** if $\mu(K) < \infty$ for every compact $K \subseteq \mathbb{R}^n$.

So the chain is: Radon = Borel regular + locally finite. Evans uses "Radon" as the default notion of "good measure on $\mathbb{R}^n$." Lebesgue measure $\mathcal{L}^n$ is the main example.

Proposition 1.10 (Inner/outer regularity)**.** *If μ is a Radon measure on $\mathbb{R}^n$, then for every measurable set A:*

1. *(Outer regularity)* $\mu(A) = \inf\{\mu(U) : U \supseteq A,\ U \text{ open}\}$.
2. *(Inner regularity)* $\mu(A) = \sup\{\mu(K) : K \subseteq A,\ K \text{ compact}\}$.

Proof: Evans pp. 5–7. Outer regularity uses Borel regularity. Inner regularity uses σ-finiteness (exhaust by balls $B(0,k)$).

I spent two hours on the inner regularity proof before seeing the trick: first prove it for finite-measure sets, then extend by writing $A = \bigcup_k (A \cap B(0,k))$ and using continuity from below. This took me a while.

1.5 Measurable functions

Definition 1.11. Let $(X, \mathcal{A})$ be a measurable space. A function $f\colon X \to [-\infty, \infty]$ is **$\mathcal{A}$-measurable** if $\{f > t\} \in \mathcal{A}$ for all $t \in \mathbb{R}$.

You could equivalently use $\{f \geq t\}$, $\{f < t\}$, or $\{f \leq t\}$. These are all the same — you get from one to the other by complements and countable intersections/unions. Evans uses the $\{f > t\}$ convention.

Proposition 1.12. *Measurable functions are closed under:*

1. *Pointwise limits: if $f_k \to f$ pointwise, each f_k measurable, then f is measurable.*
2. *Sums, products,* max, min, $|f|$.
3. *Countable* sup *and* inf*:* $\sup_k f_k$ *and* $\inf_k f_k$ *are measurable.*

Proof of (iii). $\{\sup_k f_k > t\} = \bigcup_k \{f_k > t\}$. Countable union of measurable sets. Done. The inf case is similar. ☐

Remark 1.13. Uncountable suprema can fail to be measurable. That's why pointwise limits work (sequential = countable) but arbitrary sups don't. Measurability is a countable-operations theory.

TODO: Read Evans pp. 7–14 more carefully for the approximation by simple functions stuff.

2 August 28, 2024

Question for the audience: if a sequence of measurable functions converges pointwise, is the convergence "almost uniform"? Turns out: yes, as long as you're on a finite measure space. That's Egoroff.

Today we do Egoroff and Lusin, then start on integration.

2.1 Egoroff's theorem

This says convergence of measurable functions is "almost uniform." The proof is one of those things that looks like pure epsilon management but the idea is actually pretty clean.

Theorem 2.1 (Egoroff)**.** *Let μ be a finite measure on X, and let f_k, f be measurable functions with $f_k \to f$ a.e. Then for every $\varepsilon > 0$, there exists a measurable set A with $\mu(X \setminus A) < \varepsilon$ such that $f_k \to f$ **uniformly** on A.*

Proof. For each $m, n \in \mathbb{N}$, define

$$E_{m,n} = \bigcup_{k=n}^{\infty} \{x : |f_k(x) - f(x)| \geq \tfrac{1}{m}\}.$$

Then $E_{m,n} \downarrow$ as $n \to \infty$, and the a.e. convergence gives $\mu\big(\bigcap_n E_{m,n}\big) = 0$. So by continuity from above (here we use $\mu(X) < \infty$!), for each m pick n_m with $\mu(E_{m,n_m}) < \varepsilon/2^m$. Set $A = X \setminus \bigcup_m E_{m,n_m}$. Then $\mu(X \setminus A) < \varepsilon$, and on A we have $|f_k - f| < 1/m$ for all $k \geq n_m$. That's uniform convergence. ☐

The finiteness of μ is doing real work. Counterexample: $f_k = \mathbf{1}_{[k,k+1]}$ on $\mathbb{R}$ with Lebesgue measure. Converges to 0 pointwise everywhere. But not uniformly on any set of finite complement.

TODO: check if σ-finiteness is enough, or if we really need $\mu(X) < \infty$. I think you need finiteness on the set where convergence happens.

2.2 Lusin's theorem

Theorem 2.2 (Lusin). *Let μ be a Radon measure on $\mathbb{R}^n$ and $f\colon \mathbb{R}^n \to \mathbb{R}$ measurable. Then for every $\varepsilon > 0$ and every measurable set A with $\mu(A) < \infty$, there exists a compact set $K \subseteq A$ with $\mu(A \setminus K) < \varepsilon$ such that $f|_K$ is continuous.*

Proof: Evans pp. 15–16. The idea is to use inner regularity to approximate level sets $\{a_i \leq f < a_{i+1}\}$ by compact sets. On the union of those compact sets, f is a staircase that can be refined to continuous. Tietze extension shows up at one point.

Lusin is philosophically nice — measurable functions are "almost continuous" — but I haven't needed it as much as Egoroff in practice. It comes back for the Riesz representation theorem later.

2.3 The Lebesgue integral

Evans builds the integral the standard way.

Definition 2.3. For $f\colon X \to [0,\infty]$ measurable,

$$\int_X f \,\mathrm{d}\mu = \sup\left\{\int_X s \,\mathrm{d}\mu : 0 \leq s \leq f,\ s \text{ simple}\right\},$$

where for a simple function $s = \sum_{i=1}^N a_i \mathbf{1}_{A_i}$ (with the A_i measurable, disjoint), $\int_X s \,\mathrm{d}\mu = \sum_{i=1}^N a_i\, \mu(A_i)$.

For general measurable f, write $f = f^+ - f^-$ and define $\int f \,\mathrm{d}\mu = \int f^+ \,\mathrm{d}\mu - \int f^- \,\mathrm{d}\mu$, provided at least one side is finite. If both are finite, f is **integrable** ($f \in L^1$). Nothing fancy.

2.4 Monotone convergence theorem

This is the foundation. DCT and Fatou both come from this.

Theorem 2.4 (Monotone Convergence). *Let $0 \leq f_1 \leq f_2 \leq \cdots$ be measurable with $f_k \nearrow f$ pointwise. Then*

$$\int_X f \,\mathrm{d}\mu = \lim_{k\to\infty} \int_X f_k \,\mathrm{d}\mu.$$

Proof. The $\leq$ direction: $f_k \leq f$ implies $\int f_k \leq \int f$, so $\lim \int f_k \leq \int f$.

For $\geq$: pick any simple s with $0 \leq s \leq f$ and fix $0 < t < 1$. Define $A_k = \{x : f_k(x) \geq t \cdot s(x)\}$. Then $A_k \nearrow X$ (since $f_k \nearrow f \geq s > ts$ wherever $s > 0$). So

$$\int_X f_k \,\mathrm{d}\mu \geq \int_{A_k} f_k \,\mathrm{d}\mu \geq t \int_{A_k} s \,\mathrm{d}\mu.$$

Since s is simple, $\int_{A_k} s \,\mathrm{d}\mu \to \int_X s \,\mathrm{d}\mu$ by continuity from below. So $\lim \int f_k \geq t \int s$. Let $t \nearrow 1$, then sup over simple $s \leq f$. $\square$

I initially tried to use DCT to prove this. Circular. MCT has to come first — it's the foundation that everything else rests on.

2.5 Fatou's lemma

Lemma 2.5 (Fatou)**.** *If $f_k \geq 0$ are measurable, then*

$$\int_X \liminf_{k\to\infty} f_k \, \mathrm{d}\mu \leq \liminf_{k\to\infty} \int_X f_k \, \mathrm{d}\mu.$$

Proof. Set $g_k = \inf_{j\geq k} f_j$. Then $g_k \nearrow \liminf f_k$ and $g_k \leq f_k$, so $\int g_k \leq \int f_k$. Apply MCT to the left side. □

Short and clean. The inequality can be strict — $f_k = k \cdot \mathbf{1}_{(0,1/k)}$ on $\mathbb{R}$: $\int f_k = 1$ for all k, but $\liminf f_k = 0$ a.e.

2.6 Dominated convergence

Theorem 2.6 (Dominated Convergence)**.** *Let $f_k \to f$ a.e., with $|f_k| \leq g$ a.e. for some integrable g. Then f is integrable and*

$$\lim_{k\to\infty} \int_X f_k \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu.$$

Proof. The trick is to apply Fatou twice. $g + f_k \geq 0$ and $g - f_k \geq 0$, so:

$$\int (g+f) \leq \liminf \int (g + f_k), \qquad \int (g-f) \leq \liminf \int (g - f_k).$$

Since $\int g < \infty$, cancel:

$$\int f \leq \liminf \int f_k, \qquad -\int f \leq -\limsup \int f_k.$$

So $\limsup \int f_k \leq \int f \leq \liminf \int f_k$. We're done. □

The "apply Fatou twice" move is one of those things that's obvious after you've seen it and baffling before. The domination hypothesis $|f_k| \leq g$ is doing all the heavy lifting: it makes the Fatou inequalities go the right way and keeps everything integrable.

Remark 2.7. You can't drop the domination. Same counterexample: $f_k = k \cdot \mathbf{1}_{(0,1/k)}$ converges to 0 pointwise but $\int f_k = 1$. No integrable dominator exists.

3 September 4, 2024

I accidentally tried to go to class on Monday. Labor Day. Nobody told me.

Today: product measures, Fubini–Tonelli, and then the covering theorems that set up the second half of the chapter.

3.1 Product measures and Fubini's theorem

Definition 3.1 (Product σ-algebra)**.** If $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ are measurable spaces, the **product σ-algebra** $\mathcal{A} \otimes \mathcal{B}$ on $X \times Y$ is generated by measurable rectangles $A \times B$ ($A \in \mathcal{A}$, $B \in \mathcal{B}$).

Theorem 3.2 (Product measure)**.** *If μ and ν are σ-finite measures, there exists a unique measure $\mu \otimes \nu$ on $\mathcal{A} \otimes \mathcal{B}$ with $(\mu \otimes \nu)(A \times B) = \mu(A) \cdot \nu(B)$.*

σ-finiteness is needed for uniqueness. Without it you can cook up weird examples.

Theorem 3.3 (Fubini–Tonelli)**.** *Let μ, ν be σ-finite measures.*

1. ***(Tonelli)*** *If $f \geq 0$ is $\mathcal{A} \otimes \mathcal{B}$-measurable, then*

$$\int_{X\times Y} f \,\mathrm{d}(\mu \otimes \nu) = \int_X \left(\int_Y f(x,y)\,\mathrm{d}\nu(y) \right) \mathrm{d}\mu(x) = \int_Y \left(\int_X f(x,y)\,\mathrm{d}\mu(x) \right) \mathrm{d}\nu(y).$$

2. ***(Fubini)*** *If $f \in L^1(\mu \otimes \nu)$, the same holds, and the inner integrals are finite for a.e. value of the outer variable.*

The proof is a slog but the strategy is clean: verify for indicators of measurable rectangles (trivial), extend to simple functions by linearity, extend to nonneg measurable by MCT, extend to integrable by splitting $f = f^+ - f^-$.

The point is that Tonelli is for $f \geq 0$ and doesn't need integrability, while Fubini needs $f \in L^1$. Common trick: use Tonelli on $|f|$ to check integrability, then use Fubini on f.

3.1.1 Lebesgue measure

Definition 3.4. n-dimensional **Lebesgue measure** $\mathcal{L}^n$ is the product $\underbrace{\mathcal{L}^1 \otimes \cdots \otimes \mathcal{L}^1}_{n}$, where $\mathcal{L}^1$ is the completion of the Borel measure on $\mathbb{R}$ assigning to each interval $[a,b]$ its length $b-a$.

Evans actually constructs $\mathcal{L}^n$ via the outer measure

$$\mathcal{L}^n_*(A) = \inf\left\{ \sum_{k=1}^{\infty} |Q_k| : A \subseteq \bigcup_{k=1}^{\infty} Q_k,\ Q_k \text{ cubes} \right\},$$

where $|Q|$ is the volume of the cube. Carathéodory then gives the σ-algebra and measure. The two constructions agree on Borel sets.

Fact 3.5. *$\mathcal{L}^n$ is:*

1. *A Radon measure.*
2. *Translation invariant: $\mathcal{L}^n(A + x) = \mathcal{L}^n(A)$.*
3. *Scales correctly: $\mathcal{L}^n(rA) = r^n \mathcal{L}^n(A)$ for $r > 0$.*
4. *Uniquely characterized (up to scalar) by translation invariance on Borel sets.*

3.2 Vitali's covering theorem

OK now we get into the technical stuff. Covering theorems aren't glamorous but you literally can't differentiate measures without them. I found this section painful on first reading.

Theorem 3.6 (Vitali)**.** *Let $\mathcal{F}$ be a collection of closed balls in $\mathbb{R}^n$ with $\sup\{\operatorname{diam}(B) : B \in \mathcal{F}\} < \infty$. Then there exists a countable subcollection of **disjoint** balls $\{B_i\}_{i=1}^{\infty} \subseteq \mathcal{F}$ such that*

$$\bigcup_{B\in\mathcal{F}} B \subseteq \bigcup_{i=1}^{\infty} \hat{B}_i,$$

where $\hat{B}_i$ is the ball with the same center as B_i but 5 times the radius.

The "$5r$-trick." The proof is a greedy algorithm: pick the biggest ball (or close to it), throw out everything it intersects, repeat. Why 5? Because if two balls overlap and one has radius at most the other's, the smaller one sits inside the $5\times$ enlargement. Nothing deeper than that.

There's also a refined version for Radon measures:

Theorem 3.7 (Vitali covering theorem for Radon measures)**.** *Let μ be a Radon measure, $A \subseteq \mathbb{R}^n$, and $\mathcal{F}$ a collection of closed balls such that for each $x \in A$, there exist balls in $\mathcal{F}$ centered at x with arbitrarily small radius (a "fine cover"). Then there exist disjoint balls $\{B_i\} \subseteq \mathcal{F}$ with*

$$\mu\left(A \setminus \bigcup_{i=1}^{\infty} B_i\right) = 0.$$

So a fine Vitali cover lets you cover μ-almost all of A with disjoint balls. Powerful.

3.3 Besicovitch's covering theorem

Theorem 3.8 (Besicovitch)**.** *There exists a constant N_n (depending only on dimension n) such that: if $A \subseteq \mathbb{R}^n$ is bounded and $\mathcal{F}$ is a collection of closed balls with a ball centered at each point of A, then there exist at most N_n countable subcollections $\mathcal{G}_1, \ldots, \mathcal{G}_{N_n}$, each consisting of disjoint balls, such that*

$$A \subseteq \bigcup_{j=1}^{N_n} \bigcup_{B \in \mathcal{G}_j} B.$$

Besicovitch is harder than Vitali. The proof is genuinely involved — Evans devotes pp. 30–35 to it. The advantage over Vitali: the balls don't get enlarged. The trade-off: you get bounded overlap (N_n families) instead of disjointness in one family. The constant N_n depends only on dimension, and its exact value doesn't matter.

The geometric core of the proof is a counting argument: at any point, the number of balls of comparable radius containing that point is bounded by a dimensional constant. The proof is annoying but mechanical.

I won't reproduce it. I followed it once in lecture and I don't think I could reproduce it cold.

TODO: redo the Besicovitch proof more carefully.

4 September 9, 2024

Evans derives Radon–Nikodym as a *consequence* of measure differentiation, which is the reverse of the usual functional-analysis approach. I think it's more geometric and more natural for what follows (BV functions, sets of finite perimeter), but it makes this part of the chapter harder. Worth it though.

4.1 Differentiation of Radon measures

This is where the covering theorems pay off. The goal: given two Radon measures μ and ν, differentiate ν with respect to μ by looking at ratios on shrinking balls.

Definition 4.1. For Radon measures μ, ν on $\mathbb{R}^n$, define

$$D_\mu \nu(x) = \lim_{r \to 0} \frac{\nu(B(x,r))}{\mu(B(x,r))},$$

when the limit exists (and $\mu(B(x,r)) > 0$ for small r).

Here's what makes this all work:

Theorem 4.2 (Lebesgue–Besicovitch Differentiation)**.** *Let μ be a Radon measure on $\mathbb{R}^n$ and ν a (signed) Radon measure. Then:*

1. *$D_\mu\nu(x)$ exists and is finite μ-a.e.*
2. *The Lebesgue decomposition $\nu = \nu_{ac} + \nu_s$ (where $\nu_{ac} \ll \mu$ and $\nu_s \perp \mu$) satisfies $D_\mu\nu(x) = \frac{d\nu_{ac}}{d\mu}(x)$ μ-a.e.*
3. *$D_\mu\nu_s(x) = 0$ for μ-a.e. x.*

Part (iii) is saying the singular part is "invisible" to the derivative. I still find this proof non-obvious even after going through it twice.

Proof sketch. Evans constructs the Lebesgue decomposition *using* the derivative. Define $D^+ = \limsup_{r\to 0}$ and $D_- = \liminf_{r\to 0}$ of the ratio $\nu(B(x,r))/\mu(B(x,r))$. The main technical step: show that

$$\mu\{x : D^+\nu(x) > t > s > D_-\nu(x)\} = 0$$

for all $t > s$. This uses Besicovitch to build efficient covers at two different scales. Once you know $D^+ = D_-$ a.e., the limit exists. The identification with the Radon–Nikodym derivative is then a uniqueness argument. See Evans pp. 39–43. $\square$

Corollary 4.3 (Lebesgue decomposition)**.** *Every signed Radon measure ν has a unique decomposition $\nu = \nu_{ac} + \nu_s$ with $\nu_{ac} \ll \mu$ and $\nu_s \perp \mu$.*

Corollary 4.4 (Radon–Nikodym)**.** *If $\nu \ll \mu$ (both Radon on $\mathbb{R}^n$), then there exists $f \in L^1_{\text{loc}}(\mu)$ with $\nu = f\mu$, i.e., $\nu(A) = \int_A f \,\mathrm{d}\mu$ for all Borel A. And $f = D_\mu\nu$ a.e.*

4.2 Lebesgue points

Definition 4.5 (Lebesgue point)**.** Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. A point x is a **Lebesgue point** of f if

$$\lim_{r\to 0} \frac{1}{\mathcal{L}^n(B(x,r))} \int_{B(x,r)} |f(y) - f(x)| \,\mathrm{d}y = 0.$$

This is stronger than saying the averages converge to $f(x)$. It's saying the L^1 oscillation around x vanishes. The distinction matters.

Theorem 4.6. *If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then $\mathcal{L}^n$-a.e. x is a Lebesgue point of f.*

Proof idea. Apply Lebesgue–Besicovitch (Theorem 4.2) to $\nu(A) = \int_A |f(y) - c| \,\mathrm{d}y$ for each rational c. This gives

$$\lim_{r\to 0} \frac{1}{\mathcal{L}^n(B(x,r))} \int_{B(x,r)} |f(y) - c| \,\mathrm{d}y = |f(x) - c| \quad \text{a.e.}$$

for each rational c. Take a countable intersection (one for each rational), then choose $c = c_k \to f(x)$ along rationals. You get the full Lebesgue point condition. $\square$

4.3 Approximate continuity

Definition 4.7. A measurable function f is **approximately continuous** at x if there exists a measurable set A with density 1 at x (meaning $\lim_{r\to 0} \mathcal{L}^n(A \cap B(x,r))/\mathcal{L}^n(B(x,r)) = 1$) such that $f|_A$ is continuous at x.

Theorem 4.8. *If $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$, then f is approximately continuous at $\mathcal{L}^n$-a.e. x.*

This follows from the Lebesgue point theorem: if x is a Lebesgue point, take $A = \{y : |f(y) - f(x)| < \varepsilon\}$ and check it has density 1. Details on Evans p. 46.

These concepts — Lebesgue points, approximate limits, approximate continuity — come back in a big way in Chapters 5 and 6 when Evans discusses BV functions. For now I'm just filing them away.

5 September 11, 2024

Last lecture on Chapter 1. We finish with the Riesz representation theorem and weak* compactness for measures. This connects the measure theory to functional analysis, and honestly this section feels the most "structural" compared to the covering-theorem grind from last week.

5.1 Riesz representation theorem

This next result is why we built all the Radon measure machinery.

Theorem 5.1 (Riesz Representation)**.** *Let $L\colon C_c(\mathbb{R}^n) \to \mathbb{R}$ be a positive linear functional (i.e., $L(f) \geq 0$ whenever $f \geq 0$). Then there exists a unique Radon measure μ on $\mathbb{R}^n$ such that*

$$L(f) = \int_{\mathbb{R}^n} f \, \mathrm{d}\mu \quad \textit{for all } f \in C_c(\mathbb{R}^n).$$

Here $C_c(\mathbb{R}^n)$ is continuous functions with compact support.

Proof outline. **Step 1.** Define μ on open sets by

$$\mu(U) = \sup\{L(f) : f \in C_c(\mathbb{R}^n),\ 0 \leq f \leq 1,\ \mathrm{spt}(f) \subseteq U\}.$$

Step 2. Extend to an outer measure via $\mu^*(A) = \inf\{\mu(U) : U \supseteq A,\ U \text{ open}\}$.

Step 3. Carathéodory gives a measure. Show it's Radon.

Step 4. Show $L(f) = \int f \, \mathrm{d}\mu$ by approximating f from below with simple functions built from the sets $\{f > t\}$.

Step 5. Uniqueness: if two Radon measures agree on C_c, they agree on open sets by inner regularity, hence on all Borel sets. ☐

I won't pretend I've internalized every detail. The construction in Step 1 is the key idea — the linear functional "tells you" what the measure of each open set should be, and you bootstrap from there. Actually, let me back up. The reason this works is that open sets are determined from below by continuous functions with compact support. That's the bridge.

TODO: work through the proof more carefully, maybe try it for $\mathbb{R}^1$ first.

Remark 5.2. There's a signed version: every bounded linear functional on $C_0(\mathbb{R}^n)$ (continuous functions vanishing at infinity, with the sup norm) is represented by a signed Radon measure. Evans doesn't state this but it's standard (Rudin's *Real and Complex Analysis*, Chapter 6).

5.2 Weak* convergence

Definition 5.3. A sequence of Radon measures μ_k converges **weakly*** to μ (written $\mu_k \overset{*}{\rightharpoonup} \mu$) if

$$\int_{\mathbb{R}^n} f \, \mathrm{d}\mu_k \to \int_{\mathbb{R}^n} f \, \mathrm{d}\mu \quad \text{for all } f \in C_c(\mathbb{R}^n).$$

Evans calls this "weak convergence" but it's really weak* — measures live in $(C_c)^*$, not in a space that's itself a dual. This is standard confusion in the literature and it used to trip me up.

Theorem 5.4 (Weak* compactness). *Let $\{\mu_k\}$ be Radon measures on $\mathbb{R}^n$ with $\sup_k \mu_k(K) < \infty$ for every compact $K \subseteq \mathbb{R}^n$. Then there exists a subsequence $\mu_{k_j} \overset{*}{\rightharpoonup} \mu$ for some Radon measure μ.*

Proof sketch. $C_c(\mathbb{R}^n)$ is separable. Let $\{f_m\}$ be a countable dense subset. For each m, the sequence $\{\int f_m \, \mathrm{d}\mu_k\}$ is bounded by the uniform bound on compact sets plus compact support of f_m. Diagonalize: extract a subsequence so that $\int f_m \, \mathrm{d}\mu_{k_j}$ converges for every m.

By density, $\int f \, \mathrm{d}\mu_{k_j}$ converges for every $f \in C_c$. This limit defines a positive linear functional, so by Riesz (Theorem 5.1) it's integration against a Radon measure. □

So Riesz is doing real work here — it converts the convergent functional back into a measure.

Proposition 5.5 (Lower semicontinuity). *If $\mu_k \overset{*}{\rightharpoonup} \mu$ and $U \subseteq \mathbb{R}^n$ is open, then $\mu(U) \leq \liminf_{k\to\infty} \mu_k(U)$.*

Proof. Approximate $\mathbf{1}_U$ from below by $f \in C_c$ with $0 \leq f \leq 1$, $\mathrm{spt}(f) \subseteq U$. Then $\int f \, \mathrm{d}\mu = \lim \int f \, \mathrm{d}\mu_k \leq \liminf \mu_k(U)$. Take sup over such f. □

This is one direction of the Portmanteau theorem. The other directions — $\limsup \mu_k(C) \leq \mu(C)$ for closed C, convergence of $\mu_k(A)$ when $\mu(\partial A) = 0$ — follow by similar approximation.

Example 5.6. $\mu_k = \delta_{1/k}$ on $\mathbb{R}$. Then $\mu_k \overset{*}{\rightharpoonup} \delta_0$ since $f(1/k) \to f(0)$ for all $f \in C_c(\mathbb{R})$. But $\mu_k(\{0\}) = 0$ for all k while $\delta_0(\{0\}) = 1$. Weak convergence doesn't give convergence of measures of specific sets unless the boundary has measure zero.

Remark 5.7. In probability, this compactness result is basically Prokhorov's theorem. The condition $\sup_k \mu_k(K) < \infty$ is related to tightness. For probability measures on $\mathbb{R}^n$, tightness and relative sequential compactness in the weak topology are the same thing.

That wraps up Chapter 1. A lot of machinery, but the payoff comes later when Evans can freely differentiate measures, pass to weak limits, and decompose into absolutely continuous and singular parts without reproof. The covering theorems felt like the hardest part; the Riesz representation and weak compactness feel the most useful going forward.

TODO: go back and do more examples for the differentiation section. I understand Lebesgue–Besicovitch's statement but the proof still feels fuzzy. Maybe try it for $\mathbb{R}^1$ where Besicovitch is basically just Vitali.

6 September 16, 2024

Brian asked me over lunch what Hausdorff measure even *is*, and my attempt to explain it convinced me I didn't understand the definition as well as I thought. So let's be careful.

The big idea: Lebesgue measure only "sees" n-dimensional things in $\mathbb{R}^n$. If you want to measure a curve in $\mathbb{R}^3$, or the surface of a sphere, or a fractal, you need something that can handle non-integer dimensions. That's Hausdorff measure.

6.1 Hausdorff pre-measure and Hausdorff measure

We build $\mathcal{H}^s$ in two steps: approximate at scale δ, then take a limit.

Definition 6.1 (Hausdorff pre-measure)**.** Let $A \subseteq \mathbb{R}^n$, $0 \le s < \infty$, $0 < \delta \le \infty$. Define

$$\mathcal{H}^s_\delta(A) = \inf\left\{ \sum_{j=1}^{\infty} \alpha(s)\left(\frac{\operatorname{diam} C_j}{2}\right)^s : A \subseteq \bigcup_{j=1}^{\infty} C_j,\ \operatorname{diam} C_j \le \delta \right\},$$

where $\alpha(s) = \pi^{s/2}/\Gamma(s/2+1)$ is the normalization constant chosen so that $\mathcal{H}^n$ agrees with $\mathcal{L}^n$.

I keep forgetting the normalization. Writing it down one more time: $\alpha(n) = \pi^{n/2}/\Gamma(n/2+1)$. For integer n this is the volume of the unit ball in $\mathbb{R}^n$. So $\alpha(1) = 2$, $\alpha(2) = \pi$, $\alpha(3) = 4\pi/3$.

Definition 6.2 (s-dimensional Hausdorff measure)**.**

$$\mathcal{H}^s(A) = \lim_{\delta \to 0^+} \mathcal{H}^s_\delta(A) = \sup_{\delta > 0} \mathcal{H}^s_\delta(A).$$

The limit exists (possibly $= +\infty$) because $\mathcal{H}^s_\delta(A)$ is non-decreasing as $\delta \to 0$: shrinking δ restricts the admissible covers.

So as δ shrinks, we force covers to use smaller sets, which can only make the infimum larger. Clean.

Remark 6.3. Some sanity checks:

- $\mathcal{H}^0$ is counting measure. Each nonempty covering set contributes $\alpha(0)(\operatorname{diam} C_j/2)^0 = 1$.
- $\mathcal{H}^1$ on $\mathbb{R}^1$ agrees with $\mathcal{L}^1$. Not hard but takes a small argument — intervals are the most efficient covers on the real line.
- $\mathcal{H}^n$ on $\mathbb{R}^n$ agrees with $\mathcal{L}^n$. That's the content of the isodiametric inequality below.
- $\mathcal{H}^s \equiv 0$ on $\mathbb{R}^n$ for $s > n$. Cover by cubes of side δ; each has diameter $\delta\sqrt{n}$. The sum goes to 0 as $\delta \to 0$ because the exponent $s > n$ beats the count.

$\mathcal{H}^s$ is a Borel regular outer measure on $\mathbb{R}^n$. Countable subadditivity is straightforward. Borel regularity takes more work (Evans, p. 62).

6.2 Hausdorff dimension

Lemma 6.4 (Dimension threshold)**.** *Let $A \subseteq \mathbb{R}^n$.*

1. *If $\mathcal{H}^s(A) < \infty$, then $\mathcal{H}^t(A) = 0$ for all $t > s$.*
2. *If $\mathcal{H}^t(A) > 0$, then $\mathcal{H}^s(A) = +\infty$ for all $0 \le s < t$.*

Proof sketch. For (i): let $\{C_j\}$ be a δ-cover of A. Then

$$\sum_j \alpha(t)\left(\frac{\operatorname{diam} C_j}{2}\right)^t \le \left(\frac{\delta}{2}\right)^{t-s} \frac{\alpha(t)}{\alpha(s)} \sum_j \alpha(s)\left(\frac{\operatorname{diam} C_j}{2}\right)^s.$$

Take the infimum: $\mathcal{H}^t_\delta(A) \le C \cdot \delta^{t-s} \cdot \mathcal{H}^s_\delta(A)$. Send $\delta \to 0$. The δ^{t-s} factor kills everything since $t - s > 0$ and $\mathcal{H}^s(A) < \infty$. Part (ii) is the contrapositive. □

So there's a critical value of s where $\mathcal{H}^s$ jumps from $+\infty$ to 0. That value is the Hausdorff dimension.

Definition 6.5 (Hausdorff dimension).

$$\dim_{\mathcal{H}}(A) = \inf\big\{s \geq 0 : \mathcal{H}^s(A) = 0\big\} = \sup\big\{s \geq 0 : \mathcal{H}^s(A) = +\infty\big\}.$$

At the critical dimension itself, $\mathcal{H}^s(A)$ can be 0, ∞, or anything in between. The middle-thirds Cantor set has $\dim_{\mathcal{H}} = \log 2/\log 3$ and its Hausdorff measure at that dimension is 1. That's special — most sets aren't so nice.

Some examples:

- Smooth k-dimensional submanifolds of $\mathbb{R}^n$: $\dim_{\mathcal{H}} = k$.
- Middle-thirds Cantor set: $\dim_{\mathcal{H}} = \log 2/\log 3 \approx 0.631$.
- Countable sets: $\dim_{\mathcal{H}} = 0$.
- $\mathbb{R}^n$: $\dim_{\mathcal{H}} = n$. Obviously.

I tried computing $\mathcal{H}^{\log 2/\log 3}$ of the Cantor set directly but got confused by the covering argument. The upper bound is easy (use the 2^k remaining intervals at level k), but the lower bound needs more care — you have to show those "natural" covers are actually optimal. I don't really get the intuition yet.

TODO: work through the Cantor set dimension calculation carefully. Check Rogers's book for the density argument.

7 September 18, 2024

The highlight of this chapter. The isodiametric inequality via Steiner symmetrization is one of the cleanest proofs I've seen all semester. I didn't see it coming at all.

7.1 Steiner symmetrization

Definition 7.1 (Steiner symmetrization). Let $A \subseteq \mathbb{R}^n$ be Borel and $a \in S^{n-1}$ a unit vector. The *Steiner symmetrization* of A with respect to the hyperplane $\{x : x \cdot a = 0\}$ is obtained by replacing, for each line parallel to a, the intersection of that line with A by a symmetric interval of the same length centered on the hyperplane.

Precisely: decompose $x = y + ta$ where $y \perp a$, $t \in \mathbb{R}$. For each y, let $A_y = \{t : y + ta \in A\}$. Then

$$S_a(A) = \big\{\, y + ta : y \perp a,\ |t| \leq \tfrac{1}{2}\mathcal{L}^1(A_y) \,\big\}.$$

Picture: you slide each vertical "slice" of A so it's centered on the hyperplane. Each slice keeps its length, so volume is preserved. But the set gets "rounder," so the diameter can only shrink. That's the whole game.

Lemma 7.2. *Let $A \subseteq \mathbb{R}^n$ be Borel.*

1. $\operatorname{diam} S_a(A) \leq \operatorname{diam} A$.
2. $\mathcal{L}^n(S_a(A)) = \mathcal{L}^n(A)$.

Proof sketch. Part (ii) is immediate from Fubini: each slice A_y gets replaced by an interval of the same $\mathcal{L}^1$-measure. Integrate over y.

Part (i) is the interesting one. Take $y_1 + t_1 a$ and $y_2 + t_2 a$ in $S_a(A)$. We need their distance $\leq \operatorname{diam} A$. By the symmetrization construction, there exist s_1, s_2 with $y_1 + s_1 a \in A$ and $y_2 + s_2 a \in A$, and you can choose them so that $|s_1 - s_2| \geq |t_1 - t_2|$ (because the original slices are at least as long as the symmetrized intervals, so you can pick points that are at least as far apart in the a-direction). Then:

$$|(y_1 + t_1 a) - (y_2 + t_2 a)|^2 = |y_1 - y_2|^2 + (t_1 - t_2)^2 \leq |y_1 - y_2|^2 + (s_1 - s_2)^2 \leq (\operatorname{diam} A)^2.$$

The point is that centering the intervals can only bring points in the a-direction closer together. □

7.2 The isodiametric inequality

Theorem 7.3 (Isodiametric inequality)**.** *For any Borel set $A \subseteq \mathbb{R}^n$,*

$$\mathcal{L}^n(A) \le \alpha(n)\left(\frac{\text{diam } A}{2}\right)^n.$$

Among all sets of a given diameter, the ball has the largest volume.

Proof. We can assume $\operatorname{diam} A < \infty$ (otherwise the right side is $+\infty$).

Step 1: Successive symmetrization. Let $e_1, \dots, e_n$ be the standard basis. Set $A_0 = A$ and $A_k = S_{e_k}(A_{k-1})$ for $k = 1, \dots, n$. By Lemma 7.2:

$$\mathcal{L}^n(A_n) = \mathcal{L}^n(A), \qquad \operatorname{diam} A_n \le \operatorname{diam} A.$$

And A_n is symmetric about the origin — symmetric with respect to each coordinate hyperplane.

Step 2. Since A_n is origin-symmetric, for every $x \in A_n$ we have $-x \in A_n$. So

$$2|x| = |x - (-x)| \le \operatorname{diam} A_n \le \operatorname{diam} A.$$

This gives $A_n \subseteq B(0, \operatorname{diam} A/2)$.

Step 3.

$$\mathcal{L}^n(A) = \mathcal{L}^n(A_n) \le \mathcal{L}^n\big(B(0, \operatorname{diam} A/2)\big) = \alpha(n)\left(\frac{\operatorname{diam} A}{2}\right)^n.$$

□

Symmetrize along each axis. The set becomes origin-symmetric without changing volume or increasing diameter, so it fits in a ball of radius $\operatorname{diam}/2$. Really clean.

Remark 7.4. The theorem extends to Lebesgue measurable sets and even arbitrary sets (via outer regularity — every set sits inside a Borel set of the same outer measure).

7.3 $\mathcal{H}^n = \mathcal{L}^n$

This is what all the machinery was for.

Theorem 7.5. *On $\mathbb{R}^n$, $\mathcal{H}^n = \mathcal{L}^n$.*

Proof. Two inequalities.

$\mathcal{H}^n \ge \mathcal{L}^n$: Let $\{C_j\}$ be any covering of A with $\operatorname{diam} C_j \le \delta$. By the isodiametric inequality,

$$\mathcal{L}^n(A) \le \sum_j \mathcal{L}^n(C_j) \le \sum_j \alpha(n)\left(\frac{\operatorname{diam} C_j}{2}\right)^n.$$

Take the infimum over covers: $\mathcal{L}^n(A) \le \mathcal{H}^n_\delta(A)$ for all δ, so $\mathcal{L}^n(A) \le \mathcal{H}^n(A)$.

$\mathcal{H}^n \le \mathcal{L}^n$: Cover A by cubes of side $\delta/\sqrt{n}$ (so diameter $\le \delta$). The sum $\sum_j \alpha(n)(\operatorname{diam} C_j/2)^n$ is at most $\mathcal{L}^n(A) + \varepsilon$ if you choose the cover carefully using the definition of Lebesgue outer measure. Send $\varepsilon \to 0$: $\mathcal{H}^n_\delta(A) \le \mathcal{L}^n(A)$, so $\mathcal{H}^n(A) \le \mathcal{L}^n(A)$.

The isodiametric inequality does the heavy lifting in the first direction. The second direction is basically just checking the normalization works out. □

Brian and I spent a whole afternoon on why $\mathcal{H}^n = \mathcal{L}^n$ isn't obvious. The short answer: "cover by balls" and "cover by arbitrary sets" give the same outer measure only after you prove a non-trivial geometric inequality. That's the isodiametric inequality. Without it you're stuck.

7.4 Densities

Definition 7.6 (Upper and lower s-dimensional densities)**.** Let $\mu = \mathcal{H}^s \llcorner A$. For $x \in \mathbb{R}^n$, define

$$\Theta^{*s}(\mu, x) = \limsup_{r \to 0^+} \frac{\mu(B(x,r))}{\alpha(s)\, r^s},$$
$$\Theta_*^s(\mu, x) = \liminf_{r \to 0^+} \frac{\mu(B(x,r))}{\alpha(s)\, r^s}.$$

When the limit exists, $\Theta^s(\mu,x) = \Theta^{*s}(\mu,x) = \Theta_*^s(\mu,x)$ is the *s-dimensional density* of μ at x.

The density measures how uniformly s-dimensional a set looks at small scales near x. If A is a smooth k-dimensional submanifold, then $\Theta^k(\mathcal{H}^k \llcorner A, x) = 1$ at every $x \in A$ — locally the set looks like a flat k-plane, filling the ball at the same rate as $\mathbb{R}^k$. That's the clean case.

Theorem 7.7 (Density bounds)**.** *Let $A \subseteq \mathbb{R}^n$ with $\mathcal{H}^s(A) < \infty$. Then:*

1. $\Theta^{*s}(\mathcal{H}^s \llcorner A,\, x) = 0$ *for $\mathcal{H}^s$-a.e. $x \notin A$.*
2. $2^{-s} \le \Theta^{*s}(\mathcal{H}^s \llcorner A,\, x) \le 1$ *for $\mathcal{H}^s$-a.e. $x \in A$.*

The upper bound $\Theta^{*s} \le 1$ is the easier direction — it's basically a covering argument using the definition of $\mathcal{H}^s$. The lower bound $2^{-s} \le \Theta^{*s}$ is harder and relies on a Vitali-type covering lemma for Hausdorff measure.

Remark 7.8. For integer $s = n$ everything is nicer. The n-dimensional density of $\mathcal{L}^n$-measurable sets equals 1 a.e. on the set and 0 a.e. off it — that's the Lebesgue density theorem. For non-integer s the density may not exist at all, and the gap between 2^{-s} and 1 is genuine. Sets with non-integer Hausdorff dimension don't have clean local structure. This definition felt arbitrary until I saw how it shows up in rectifiability later.

TODO: understand the proof of the lower bound 2^{-s} better. Evans uses a covering argument (pp. 72–73) that I need to go through line by line.

7.5 Connection to functions

The last bit connects Hausdorff measure back to the "fine properties" theme of the book. The main tool is Fubini applied to subgraphs.

Definition 7.9 (Subgraph)**.** For $f \colon \mathbb{R}^n \to [0,\infty]$ measurable, the *subgraph* is

$$\mathrm{SG}(f) = \left\{\, (x,t) \in \mathbb{R}^n \times \mathbb{R} : 0 \le t \le f(x) \,\right\} \subseteq \mathbb{R}^{n+1}.$$

Lemma 7.10. *If $f \colon \mathbb{R}^n \to [0,\infty]$ is $\mathcal{L}^n$-measurable, then* $\mathrm{SG}(f)$ *is $\mathcal{L}^{n+1}$-measurable and*

$$\mathcal{L}^{n+1}\big(\mathrm{SG}(f)\big) = \int_{\mathbb{R}^n} f \,\mathrm{d}\mathcal{L}^n.$$

Proof sketch. By Fubini: $\mathcal{L}^{n+1}(\mathrm{SG}(f)) = \int_{\mathbb{R}^n} \mathcal{L}^1(\{t : 0 \le t \le f(x)\})\,\mathrm{d}\mathcal{L}^n(x) = \int_{\mathbb{R}^n} f(x)\,\mathrm{d}\mathcal{L}^n(x)$. The measurability of $\mathrm{SG}(f)$ as a subset of $\mathbb{R}^{n+1}$ follows from approximation by simple functions. □

One of those lemmas that feels obvious but needs Fubini to make honest. The point for later: you can study integrability, level sets, and regularity of a function by looking at its subgraph as a subset of $\mathbb{R}^{n+1}$ and applying Hausdorff measure tools. The Cavalieri principle — $\int f = \int_0^\infty \mathcal{L}^n(\{f > t\})\,\mathrm{d}t$ — is a close relative.

That's Chapter 2. We've got Hausdorff measure, Hausdorff dimension, the isodiametric inequality via Steiner symmetrization, the proof that $\mathcal{H}^n = \mathcal{L}^n$, density results, and the subgraph connection. The isodiametric inequality proof is the gem. The rest is careful bookkeeping with covers and limits.

TODO: read Evans Section 2.4 more carefully for measurability details. Revisit the Vitali covering argument in the density lower bound.

8 September 30, 2024

Slept through most of the weekend trying to catch up. Hausdorff measure is still sitting in my head like an undigested meal, but we're moving on to the area and coarea formulas now. These are the real payoff of Chapters 1–2.

8.1 Lipschitz Functions

Definition 8.1 (Lipschitz function)**.** A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is **Lipschitz** if there exists a constant $L \geq 0$ such that

$$|f(x) - f(y)| \leq L|x - y| \qquad \text{for all } x, y \in \mathbb{R}^n.$$

The smallest such L is denoted $\mathrm{Lip}(f)$.

Why Lipschitz? Smooth maps are too restrictive for geometric measure theory, and continuous maps are too wild. Lipschitz maps are differentiable a.e. (Rademacher), they send null sets to null sets, and they don't blow up Hausdorff measure too badly. That's the sweet spot.

Theorem 8.2 (McShane Extension)**.** *Let $A \subset \mathbb{R}^n$ and $f : A \to \mathbb{R}$ be Lipschitz with constant L. Then there exists $\bar{f} : \mathbb{R}^n \to \mathbb{R}$ with $\bar{f} = f$ on A and* $\mathrm{Lip}(\bar{f}) = \mathrm{Lip}(f)$*.*

The extension is explicit:

$$\bar{f}(x) = \inf_{a \in A} \big\{ f(a) + L|x - a| \big\}.$$

Check it satisfies the Lipschitz bound. Check it agrees with f on A. Done. That's the whole proof.

For vector-valued $f : A \to \mathbb{R}^m$, you apply McShane component by component, but the Lipschitz constant picks up a factor of $\sqrt{m}$. Nobody cares about the exact constant in practice.

8.2 Rademacher's Theorem

This is one of those results that sounds impossible the first time you hear it.

Theorem 8.3 (Rademacher)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be locally Lipschitz. Then f is differentiable $\mathcal{L}^n$-a.e.*

Every locally Lipschitz function has a classical derivative at almost every point. No extra hypotheses needed. The proof is long but every individual step is manageable. It's the orchestration that's hard.

Proof sketch (Evans–Gariepy, pp. 81–84). It suffices to treat $f : \mathbb{R}^n \to \mathbb{R}$ (apply componentwise).

Step 1: Directional derivatives exist a.e. Fix a direction $v \in S^{n-1}$. For $\mathcal{L}^n$-a.e. x, the function $t \mapsto f(x + tv)$ is absolutely continuous on bounded intervals. This is basically Fubini: project $\mathbb{R}^n$ along v, and on almost every parallel line, f is absolutely continuous. So the derivative

$$D_v f(x) = \lim_{t \to 0} \frac{f(x + tv) - f(x)}{t}$$

exists for $\mathcal{L}^n$-a.e. x. Nothing deep here.

Step 2: A "gradient" exists a.e. Pick a countable dense set $\{v_k\} \subset S^{n-1}$. By Step 1 and a countable intersection, there's a full-measure set E where $D_{v_k} f(x)$ exists for all k. At a.e. point of E, the map $v \mapsto D_v f(x)$ turns out to be linear in v. Why? Because $D_{e_i} f$ exists a.e. for each coordinate direction e_i, and on lines we get

$$D_v f(x) = \sum_{i=1}^{n} v_i \, D_{e_i} f(x)$$

for the dense set of v's, and then for all v by continuity (the Lipschitz bound gives $|D_v f| \leq \mathrm{Lip}(f)$, so we can pass to limits). Define $\nabla f(x) = (D_{e_1} f(x), \dots, D_{e_n} f(x))$. Then $D_v f(x) = \nabla f(x) \cdot v$ for all v.

Step 3: Differentiability. We need

$$Q(x, v, t) := \frac{f(x + tv) - f(x) - \nabla f(x) \cdot (tv)}{t} \to 0$$

as $t \to 0$, *uniformly in* v. This is the hardest part. Define the "bad set"

$$A_\epsilon = \big\{ x : \limsup_{t \to 0} \sup_{|v|=1} |Q(x, v, t)| > \epsilon \big\}.$$

We want $\mathcal{L}^n(A_\epsilon) = 0$. At points where ∇f exists and equals some linear map L, compare $f(x + tv) - f(x)$ against $L \cdot (tv)$ using the Lipschitz bound. A covering argument (Vitali, from a few lectures ago) shows the set where uniform convergence fails has measure zero. The uniform bound $|Q| \leq 2\,\mathrm{Lip}(f)$ is what makes the covering argument work. □

Long proof. Worth it.

Corollary 8.4. *If $f : \mathbb{R}^n \to \mathbb{R}^m$ is Lipschitz and $f = 0$ on a set Z, then $Df = 0$ $\mathcal{L}^n$-a.e. on Z.*

This follows immediately: at a.e. point of Z where f is differentiable, the derivative has to be zero because f vanishes on Z and Z has full density at such points.

We also get a chain rule: if f and g are Lipschitz and the composition makes sense, $D(g \circ f) = Dg \circ Df$ a.e. Standard.

8.3 Linear Maps and Jacobians

Quick review of the linear algebra we need (Evans–Gariepy, pp. 87–90). Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be linear.

- The **adjoint** $L^* : \mathbb{R}^m \to \mathbb{R}^n$ satisfies $\langle Lx, y \rangle = \langle x, L^* y \rangle$. In matrices, $L^* = L^T$.
- $L^* L : \mathbb{R}^n \to \mathbb{R}^n$ is symmetric and positive semidefinite.
- **Polar decomposition**: $L = O \circ S$ where $S = (L^* L)^{1/2}$ is the unique positive semidefinite square root of $L^* L$, and $O : \mathbb{R}^n \to \mathbb{R}^m$ satisfies $|Ox| = |x|$ for $x \in \mathrm{range}(S)$. When $n \leq m$ and L is injective, O extends to an orthogonal map on all of $\mathbb{R}^n$.

Definition 8.5 (Jacobian)**.** For a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$:

- If $n \leq m$: $J_L = \sqrt{\det(L^* L)}$. This equals the product of singular values $\lambda_1 \cdots \lambda_n$ where λ_i^2 are eigenvalues of $L^* L$.

- If $n \geq m$: $J_L = \sqrt{\det(LL^*)}$.

For a Lipschitz map $f : \mathbb{R}^n \to \mathbb{R}^m$, the **Jacobian** at a point of differentiability is $J_f(x) = J_{Df(x)}$.

When $n = m$, both definitions agree and give $J_L = |\det L|$. Reassuring.

Lemma 8.6 (Change of variables for linear maps). *Let $L : \mathbb{R}^n \to \mathbb{R}^m$ be linear with $n \leq m$, and let $A \subset \mathbb{R}^n$ be Lebesgue measurable. Then*

$$\mathcal{H}^n(L(A)) = J_L \cdot \mathcal{L}^n(A).$$

So for injective linear maps, L scales n-dimensional volume by exactly J_L. The area formula generalizes this to Lipschitz maps.

8.4 The Area Formula

The next result is the whole reason we defined Lipschitz maps and Jacobians.

Theorem 8.7 (Area Formula (Evans–Gariepy, Thm. 1, p. 96)). *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be Lipschitz, with $n \leq m$. Then for any Lebesgue measurable $g : \mathbb{R}^n \to [0, \infty]$,*

$$\int_{\mathbb{R}^n} g(x)\, J_f(x)\, d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} \left(\sum_{x \in f^{-1}(\{y\})} g(x) \right) d\mathcal{H}^n(y).$$

OK so when $g = \chi_A$ for some measurable A, this becomes

$$\int_A J_f\, dx = \int_{\mathbb{R}^m} \#\big(A \cap f^{-1}(\{y\})\big)\, d\mathcal{H}^n(y),$$

which says the integral of the Jacobian over A equals the integral of the number of preimages, counted with Hausdorff measure. When f is injective on A, the right side collapses to $\mathcal{H}^n(f(A))$.

Proof sketch (pp. 96–101). Here's the architecture.

Step 1: Reduce to $g = \chi_A$. Standard approximation by simple functions.

Step 2: Handle $\{J_f = 0\}$. Where $J_f(x) = 0$, the derivative $Df(x)$ has rank $< n$, so f is crushing an n-dimensional region into something lower-dimensional. Need to show $\mathcal{H}^n(f(Z)) = 0$ for $Z = \{J_f = 0\}$. Cover Z with balls, estimate the Hausdorff measure of the image using the fact that f maps small balls into thin ellipsoids. Tedious but not conceptually hard.

Step 3: Linear case. Already done (Lemma 8.6).

Step 4: Lipschitz linearization. This is the core. By Rademacher, f is differentiable a.e. At points of differentiability, $f(y) \approx f(x) + Df(x)(y - x)$ for y near x. For any $\epsilon > 0$, decompose $\mathbb{R}^n = N \cup \bigcup_{k=1}^{\infty} E_k$ where $\mathcal{L}^n(N) = 0$ and on each E_k:

- Df is nearly constant: $\|Df(x) - Df(y)\| < \epsilon$ for $x, y \in E_k$,
- f is nearly affine: $|f(x) - f(y) - Df(z)(x - y)| < \epsilon|x - y|$ for $x, y, z \in E_k$.

On each E_k, f behaves like a linear map up to error ϵ, so the Jacobian formula holds up to $(1 + C\epsilon)$. Send $\epsilon \to 0$.

Step 5: Non-injectivity. The counting function $y \mapsto \#(A \cap f^{-1}(\{y\}))$ is measurable (this isn't obvious; it follows from a structure theorem for Lipschitz maps). The multiplicity on the right accounts for non-injectivity. □

The decomposition in Step 4 sometimes goes by “Lipschitz linearization.” It appears everywhere in GMT. The point is that Lipschitz maps are well-approximated by their derivatives, and the error is controlled uniformly on compact subsets where the derivative doesn’t vary much.

8.5 Surface area as an application

Let $U \subset \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$ a Lipschitz parametrization of a surface $\Sigma = f(U)$ with $n \leq m$. The area formula gives

$$\mathcal{H}^n(\Sigma) = \int_U J_f(x)\, dx,$$

provided f is injective (or counting multiplicity if it isn’t). For a graph $f(x) = (x, u(x))$ where $u : \mathbb{R}^n \to \mathbb{R}$, we get

$$J_f(x) = \sqrt{1 + |\nabla u(x)|^2},$$

which recovers the classical surface area formula from multivariable calculus. It’s kind of wild how much machinery goes into justifying something you learn in vector calculus as a given.

9 October 2, 2024

Midterm season. I’ve been living off coffee and the vending machine on the third floor. Today we finish Chapter 3 with the coarea formula.

9.1 The Coarea Formula

I found this harder to internalize than the area formula. Something about integrating over level sets instead of over the image makes the geometry less visual.

The area formula handles $n \leq m$: a low-dimensional domain mapping into a higher-dimensional space. The coarea formula handles $n \geq m$: a high-dimensional domain mapping *down* to a lower-dimensional space. The level sets $f^{-1}(\{y\})$ are generically $(n-m)$-dimensional, and we integrate over them.

Theorem 9.1 (Coarea Formula (Evans–Gariepy, Thm. 2, p. 112))**.** *Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be Lipschitz, with $n \geq m$. Then for any Lebesgue measurable $g : \mathbb{R}^n \to [0, \infty]$,*

$$\int_{\mathbb{R}^n} g(x)\, J_f(x)\, d\mathcal{L}^n(x) = \int_{\mathbb{R}^m} \left(\int_{f^{-1}(\{y\})} g(x)\, d\mathcal{H}^{n-m}(x) \right) dy.$$

The outer integral on the right is over $\mathbb{R}^m$ with Lebesgue measure. The inner integral is over the level set $f^{-1}(\{y\})$ with $(n-m)$-dimensional Hausdorff measure.

9.2 Why this is a nonlinear Fubini theorem

Take $n = n_1 + n_2$, $m = n_2$, and let $f : \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \to \mathbb{R}^{n_2}$ be the projection $f(x_1, x_2) = x_2$. Then Df is the projection matrix, $J_f = 1$, and each level set $f^{-1}(\{y\})$ is just $\mathbb{R}^{n_1} \times \{y\}$. The coarea formula becomes

$$\int_{\mathbb{R}^n} g(x)\, dx = \int_{\mathbb{R}^{n_2}} \left(\int_{\mathbb{R}^{n_1}} g(x_1, y)\, dx_1 \right) dy,$$

which is Fubini. So the coarea formula is a genuinely nonlinear version of Fubini’s theorem.

9.3 The case $m = 1$

This is the most important special case and I kept getting confused about it until I worked through three examples. Let $f : \mathbb{R}^n \to \mathbb{R}$ be Lipschitz. Then $J_f = |\nabla f|$. The coarea formula becomes

$$\int_{\mathbb{R}^n} g(x)\, |\nabla f(x)|\, dx = \int_{-\infty}^{\infty} \left(\int_{f^{-1}(\{t\})} g\, d\mathcal{H}^{n-1} \right) dt.$$

Taking $g = \chi_A / |\nabla f|$ where $\nabla f \neq 0$:

$$\mathcal{L}^n(A) = \int_{-\infty}^{\infty} \int_{A \cap \{f=t\}} \frac{1}{|\nabla f|}\, d\mathcal{H}^{n-1}\, dt.$$

This "slices" the set A by level sets of f and reconstructs its volume. Incredibly useful in PDE and geometric analysis.

9.4 Proof sketch of the coarea formula

Proof sketch (Evans–Gariepy, pp. 112–117). Structurally similar to the area formula proof. Same architecture, different geometry.

Step 1: Linear case. For a surjective linear map $L : \mathbb{R}^n \to \mathbb{R}^m$, verify that

$$J_L \cdot \mathcal{L}^n(A) = \int_{\mathbb{R}^m} \mathcal{H}^{n-m}(A \cap L^{-1}(\{y\}))\, dy$$

for all measurable A. The level sets of L are translates of $\ker L$, an $(n-m)$-dimensional subspace. So we're computing volume via "height times cross-section," weighted by the Jacobian. Use the polar decomposition and compute.

Step 2: Handle $\{J_f = 0\}$. Where $J_f(x) = 0$, the map $Df(x)$ isn't surjective. Show that for a.e. $y \in \mathbb{R}^m$, the set $\{J_f = 0\} \cap f^{-1}(\{y\})$ has $\mathcal{H}^{n-m}$-measure zero. This uses Sard-type estimates: $f(\{J_f = 0\})$ has $\mathcal{L}^m$-measure zero. That's Sard's theorem for Lipschitz maps, which is nontrivial.

Step 3: Lipschitz linearization. Same trick as the area formula. Decompose $\mathbb{R}^n$ into pieces where Df is nearly constant, approximate by the linear formula on each piece, control errors, take limits.

Step 4: Measurability of the sliced integral. Verify that $y \mapsto \int_{f^{-1}(\{y\})} g\, d\mathcal{H}^{n-m}$ is measurable. Follows from the theory of rectifiable sets and the structure of Lipschitz level sets. □

The proof is honestly harder than the area formula. The difficulty is in Step 2: understanding the critical set and how it interacts with level sets. Sard's theorem for smooth maps is standard, but the Lipschitz version requires more care. This proof is a slog but the result is clean.

Corollary 9.2. *Let $u : \mathbb{R}^n \to \mathbb{R}$ be Lipschitz. Then for a.e. $t \in \mathbb{R}$, the level set $\{u = t\}$ has finite $\mathcal{H}^{n-1}$-measure, and*

$$\int_{-\infty}^{\infty} \mathcal{H}^{n-1}(\{u = t\} \cap A)\, dt \leq \mathrm{Lip}(u) \cdot \mathcal{L}^n(A)$$

for every bounded measurable A.

This comes from the coarea formula with $g = \chi_A$ and the bound $|\nabla u| \le \mathrm{Lip}(u)$. It tells us that "most" level sets of a Lipschitz function are nice $(n-1)$-dimensional surfaces. The bad ones form a null set in the target. This connects directly to the theory of sets of finite perimeter later on.

I still don't have good intuition for why the coarea formula involves the Jacobian of the gradient rather than, say, the determinant of the full derivative. I think it's because the Jacobian measures how much f "spreads" in the directions transverse to the level sets, and the component of Df along the level sets doesn't contribute to the slicing. But handwaving a bit here.

10 October 7, 2024

Rain all morning. The walk to class was miserable but at least the topic is good. We're starting Sobolev spaces, which is where analysis starts feeling like it has teeth.

10.1 Weak Derivatives

Definition 10.1 (Weak derivative)**.** Let $U \subseteq \mathbb{R}^n$ be open. Suppose $u, v \in L^1_{\mathrm{loc}}(U)$. We say v is the **α-th weak partial derivative** of u, written $D^\alpha u = v$, provided

$$\int_U u\, D^\alpha \varphi\, dx = (-1)^{|\alpha|} \int_U v\, \varphi\, dx$$

for all test functions $\varphi \in C_c^\infty(U)$.

Here's what's going on. You move all the derivatives onto the smooth test function via integration by parts, then *define* the derivative of u to be whatever v makes the identity work. If u happens to be classically differentiable, you recover the usual derivative. But now functions with corners, kinks, jumps in derivatives—all fair game.

Weak derivatives are unique up to a.e. equivalence. This is the fundamental lemma of the calculus of variations, nothing more.

10.2 Sobolev Spaces

Definition 10.2 (Sobolev space $W^{k,p}(U)$)**.** For $k \in \mathbb{N}$ and $1 \le p \le \infty$, the Sobolev space $W^{k,p}(U)$ consists of all $u \in L^p(U)$ such that for each multi-index α with $|\alpha| \le k$, the weak derivative $D^\alpha u$ exists and belongs to $L^p(U)$.

The norm:

$$\|u\|_{W^{k,p}(U)} = \begin{cases} \left(\displaystyle\sum_{|\alpha| \le k} \int_U |D^\alpha u|^p\, dx \right)^{1/p} & 1 \le p < \infty, \\ \displaystyle\sum_{|\alpha| \le k} \operatorname*{ess\,sup}_U |D^\alpha u| & p = \infty. \end{cases}$$

$W^{k,p}(U)$ is a Banach space. For $p = 2$ it's a Hilbert space, usually written $H^k(U)$.

$W_0^{k,p}(U)$ is the closure of $C_c^\infty(U)$ in $W^{k,p}(U)$. Think of these as Sobolev functions that "vanish on ∂U" in some appropriate sense.

Product rules and chain rules work. If $u \in W^{1,p}(U)$ and $\Phi \in C^1$ with Φ' bounded, then $\Phi \circ u \in W^{1,p}(U)$ and $D(\Phi \circ u) = \Phi'(u) Du$.

One thing that confused me early on: $W^{k,p}(U)$ is *not* just "C^k functions completed in the $W^{k,p}$ norm." That completion starting from C_c^∞ gives you $W_0^{k,p}$, which is a different space. You genuinely need the weak derivative definition.

10.3 Approximation by Smooth Functions

The mollification proof is the template for everything in this chapter. Master it once and you're set.

Definition 10.3 (Mollifier)**.** Let $\eta \in C_c^\infty(\mathbb{R}^n)$ be the standard mollifier with $\eta \geq 0$, $\operatorname{spt}\eta \subseteq B(0,1)$, and $\int \eta = 1$. Set $\eta_\varepsilon(x) = \varepsilon^{-n}\eta(x/\varepsilon)$.

For $u \in L^1_{\text{loc}}(U)$, the mollification is $u^\varepsilon = \eta_\varepsilon * u$ on $U_\varepsilon = \{x \in U : \operatorname{dist}(x, \partial U) > \varepsilon\}$.

Theorem 10.4 (Local approximation)**.** *If $u \in W^{k,p}(U)$ with $1 \leq p < \infty$, then $u^\varepsilon \in C^\infty(U_\varepsilon)$ and $u^\varepsilon \to u$ in $W^{k,p}_{\text{loc}}(U)$ as $\varepsilon \to 0$.*

Proof sketch. The trick is showing $D^\alpha(\eta_\varepsilon * u) = \eta_\varepsilon * D^\alpha u$ for $|\alpha| \leq k$ on U_ε. Write out the convolution, differentiate under the integral (legal since η_ε is smooth and compactly supported), recognize the result as $(-1)^{|\alpha|} \int u \, D_y^\alpha \eta_\varepsilon(\cdot - y)\, dy$, then use the definition of weak derivative to flip the derivatives back. Once you have this identity, L^p convergence is the standard mollification result. □

Now the local-to-global step.

Theorem 10.5 (Meyers–Serrin)**.** *$C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$ for $1 \leq p < \infty$.*

The proof uses a partition of unity subordinate to an exhaustion of U. You mollify each piece with a small enough ε and sum. It's a careful $\varepsilon/2^k$ argument, nothing deep, but you have to track the supports.

Can you approximate by $C^\infty(\overline{U})$ functions? In general, no.

Theorem 10.6 (Approximation up to the boundary)**.** *If ∂U is Lipschitz, then $C^\infty(\overline{U})$ is dense in $W^{k,p}(U)$ for $1 \leq p < \infty$.*

The Lipschitz condition lets you locally flatten the boundary and push the function slightly inward, where you can mollify safely. Without it, you can build domains where the approximation fails.

Three approximation theorems, each stronger, each needing more hypotheses.

11 October 9, 2024

Office hours ran long yesterday so I didn't get to review traces before today. Flying a bit blind.

11.1 Traces

For smooth functions $u \in C^\infty(\overline{U})$, restricting to ∂U is trivial. For Sobolev functions you can't just "evaluate on the boundary" because they're only defined a.e. The trace operator fixes this.

Theorem 11.1 (Trace theorem)**.** *Assume U is bounded and ∂U is Lipschitz. There exists a bounded linear operator*

$$T : W^{1,p}(U) \to L^p(\partial U)$$

such that $Tu = u|_{\partial U}$ for $u \in W^{1,p}(U) \cap C(\overline{U})$, and $\|Tu\|_{L^p(\partial U)} \leq C\|u\|_{W^{1,p}(U)}$.

The trace characterizes $W_0^{1,p}(U)$: a function $u \in W^{1,p}(U)$ belongs to $W_0^{1,p}(U)$ if and only if $Tu = 0$ on ∂U. So $W_0^{1,p}$ really is the space of Sobolev functions vanishing at the boundary, exactly as the notation suggests.

The sharper statement is that T maps into the fractional Sobolev space $W^{1-1/p,p}(\partial U)$, but Evans–Gariepy don't go there.

11.2 Extensions

Theorem 11.2 (Extension). *Assume U is bounded and ∂U is Lipschitz. For each $1 \leq p \leq \infty$, there exists a bounded linear operator*

$$E : W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$$

such that $Eu = u$ a.e. in U, $\operatorname{spt}(Eu)$ is bounded, and

$$\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)}.$$

The idea: near the boundary, reflect the function across ∂U. Because ∂U is Lipschitz, you can locally flatten it, reflect, then patch with a partition of unity. Multiply by a cutoff for bounded support.

Extensions matter because lots of results are easier to prove on all of $\mathbb{R}^n$ with no boundary issues. Then you transfer back via E. The Sobolev inequalities are the prime example.

11.3 Gagliardo–Nirenberg–Sobolev ($p < n$)

This is the heart of the chapter. The notation is brutal.

The exponent $p^* = np/(n-p)$ looks arbitrary until you do dimensional analysis. $\|u\|_{L^q}$ has dimensions of $(\text{length})^{n/q}$ times pointwise size, and $\|Du\|_{L^p}$ has dimensions of $(\text{length})^{n/p-1}$ times pointwise size. For the inequality to be scale-invariant, we need $n/q = n/p - 1$, giving $q = np/(n-p) = p^*$. That's the only exponent that works.

Theorem 11.3 (Gagliardo–Nirenberg–Sobolev). *Assume $1 \leq p < n$. There exists $C = C(p, n)$ such that*

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|Du\|_{L^p(\mathbb{R}^n)}$$

for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = \dfrac{np}{n-p}$.

The right-hand side only involves Du, not u itself. That's because we're on all of $\mathbb{R}^n$ with compactly supported functions.

By density this extends to $W^{1,p}(\mathbb{R}^n)$, and via the extension operator to bounded domains with Lipschitz boundary:

$$\|u\|_{L^{p^*}(U)} \leq C\|u\|_{W^{1,p}(U)}.$$

So $W^{1,p}(U) \hookrightarrow L^{p^*}(U)$ continuously. You trade one derivative for better integrability.

Corollary 11.4 (Poincaré inequality). *If U is bounded and $1 \leq p < n$, then for $u \in W_0^{1,p}(U)$:*

$$\|u\|_{L^q(U)} \leq C\|Du\|_{L^p(U)} \quad \textit{for all } 1 \leq q \leq p^*.$$

11.4 Morrey's Inequality ($p > n$)

Morrey's inequality is what makes $p > n$ special. You get continuity for free.

Theorem 11.5 (Morrey's inequality). *Assume $n < p \leq \infty$. There exists $C = C(p, n)$ such that*

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C\|Du\|_{L^p(\mathbb{R}^n)}$$

for all $u \in C_c^1(\mathbb{R}^n)$, where $\gamma = 1 - n/p$.

So $W^{1,p}(U) \hookrightarrow C^{0,1-n/p}(\overline{U})$ for $p > n$. One derivative in L^p with p large enough, and the function *has* to be Hölder continuous. As $p \to \infty$, the Hölder exponent $\gamma \to 1$ and you approach Lipschitz. As $p \to n^+$, $\gamma \to 0$ and you just barely get continuity.

The proof integrates $|u(x) - u(y)|$ along line segments, uses polar coordinates, and hits it with Hölder's inequality. The key geometric observation: the average of $|Du|$ over a ball controls the oscillation of u.

11.5 The borderline $p = n$

The case $p = n$ is subtle. You don't get L^∞ (that would be Morrey, but $p = n$ doesn't satisfy $p > n$), and $p^* = np/(n-p)$ blows up. What you get is $W^{1,n}(U) \hookrightarrow L^q(U)$ for all $q < \infty$, but *not* $q = \infty$. Almost, but not quite.

11.6 Embedding summary

For U bounded with Lipschitz boundary:

Regime	**Embedding**	**Key feature**
$1 \le p < n$	$W^{1,p}(U) \hookrightarrow L^{p^*}(U)$	Improved integrability
$p = n$	$W^{1,n}(U) \hookrightarrow L^q(U)$, all $q < \infty$	Almost L^∞
$p > n$	$W^{1,p}(U) \hookrightarrow C^{0,1-n/p}(\overline{U})$	Hölder continuity

The dimension n is the threshold. Everything pivots on whether p is below, at, or above n.

12 October 14, 2024

I think I finally understand the Sobolev embeddings. It only took three passes through the proofs. Today we do compactness, capacity, and the fine properties of Sobolev functions.

12.1 Rellich–Kondrachov Compactness

This is Arzelà–Ascoli for Sobolev spaces.

Theorem 12.1 (Rellich–Kondrachov)**.** *Assume $U \subset \mathbb{R}^n$ is bounded and open with Lipschitz boundary. Then:*

1. *If $1 \le p < n$: $W^{1,p}(U) \hookrightarrow\hookrightarrow L^q(U)$ for each $1 \le q < p^*$.*
2. *If $p = n$: $W^{1,p}(U) \hookrightarrow\hookrightarrow L^q(U)$ for each $1 \le q < \infty$.*
3. *If $p > n$: $W^{1,p}(U) \hookrightarrow\hookrightarrow C^{0,\beta}(\overline{U})$ for each $0 < \beta < 1 - n/p$.*

Here $\hookrightarrow\hookrightarrow$ means compact embedding: bounded sequences have convergent subsequences.

Notice the strict inequalities. You get compactness into L^q for $q < p^*$, but *not* at the critical exponent $q = p^*$. The embedding $W^{1,p} \hookrightarrow L^{p^*}$ is continuous but not compact. This matters enormously in the calculus of variations.

Proof idea for case (1). Take a bounded sequence (u_k) in $W^{1,p}(U)$. Extend to $\mathbb{R}^n$, then mollify. For fixed $\varepsilon > 0$, the mollified sequence (u_k^ε) is bounded in C^1 on compact sets (from the $W^{1,p}$ bound and properties of convolution), so Arzelà–Ascoli gives a convergent subsequence. Diagonal argument over $\varepsilon \to 0$. The $q < p^*$ restriction enters because you need spare integrability to interpolate between L^1 convergence of mollifications and the L^{p^*} bound. □

I'm not going to pretend I understand every detail of the interpolation step. The mechanism is: Sobolev regularity gives equicontinuity after mollification, Arzelà–Ascoli gives convergence, interpolation gives L^q convergence for subcritical q.

12.2 Capacity

Capacity confused me for weeks. Here's why that matters: sets of capacity zero are the sets that Sobolev functions can't "see." Measure zero sets are invisible to L^p functions. Capacity zero sets are invisible to Sobolev functions. But capacity zero is a *finer* notion.

Definition 12.2 (p-capacity)**.** For $1 \le p < n$ and $A \subseteq \mathbb{R}^n$,

$$\operatorname{cap}_p(A) = \inf \left\{ \int_{\mathbb{R}^n} |Du|^p \, dx : u \in W^{1,p}(\mathbb{R}^n),\ u \ge 1 \text{ on a neighborhood of } A \right\}.$$

Properties:

1. Monotone: $A \subseteq B \implies \operatorname{cap}_p(A) \le \operatorname{cap}_p(B)$.
2. Countably subadditive: $\operatorname{cap}_p\big(\bigcup_k A_k\big) \le \sum_k \operatorname{cap}_p(A_k)$.
3. If $\operatorname{cap}_p(A) = 0$, then $\mathcal{H}^s(A) = 0$ for all $s > n - p$.
4. $\operatorname{cap}_p(A) = 0 \implies \mathcal{L}^n(A) = 0$, but not conversely.

The relationship to Hausdorff measure is the most useful one. It tells you the dimension of the exceptional set where Sobolev functions can misbehave.

12.3 Quasicontinuity and Precise Representatives

A Sobolev function is only defined a.e., so talking about pointwise values is dangerous. But you can choose a representative that's well-behaved outside an arbitrarily small set in the capacity sense.

Definition 12.3 (Quasicontinuous)**.** A function $u : U \to \mathbb{R}$ is **p-quasicontinuous** if for every $\varepsilon > 0$ there exists an open set V with $\operatorname{cap}_p(V) < \varepsilon$ such that $u|_{U \setminus V}$ is continuous.

Theorem 12.4 (Quasicontinuous representative)**.** *Every $u \in W^{1,p}(\mathbb{R}^n)$ $(1 \le p \le n)$ has a p-quasicontinuous representative $\tilde{u}$, unique up to sets of* cap_p*-zero.*

So Sobolev functions are "almost continuous." The discontinuity set can be made as small as you want in the capacity sense. Not continuous, not just measurable, but quasicontinuous. That's the right notion of regularity for these functions.

The **precise representative** is defined pointwise:

$$u^*(x) = \lim_{r \to 0} \fint_{B(x,r)} u \, dy,$$

whenever this limit exists. For $u \in W^{1,p}$, u^* exists and equals $\tilde{u}$ outside a set of cap_p-zero.

This connects back to the differentiation of measures stuff from the Vitali covering lectures. Same idea of recovering pointwise information from integral averages, but now the exceptional set is measured by capacity instead of Hausdorff measure.

12.4 Differentiability on Lines

Here's an equivalent way to think about Sobolev functions that avoids integration by parts entirely.

Theorem 12.5. *If $u \in W^{1,p}(U)$ for $1 \le p \le \infty$, then for each $i = 1, \dots, n$, the function $t \mapsto u(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_n)$ is absolutely continuous on the relevant line segment for $\mathcal{L}^{n-1}$-a.e. choice of the remaining variables. Its classical derivative equals the weak derivative $D_i u$ a.e.*

The converse is also true: if $u \in L^p(U)$ is absolutely continuous on a.e. line in each coordinate direction, and the classical partials are in L^p, then $u \in W^{1,p}(U)$.

Sobolev functions are exactly the L^p functions that are absolutely continuous on almost every axis-parallel line. That's a clean characterization. Brian asked me why you can't just use this as the definition from the start, and honestly I think you could, but the integration-by-parts version plays better with the PDE applications.

13 October 21, 2024

Midterm grades came back. Could've been worse. Today we start Chapter 5, which is about BV functions and sets of finite perimeter. Evans pp. 166–226 is one of the hardest stretches in the book.

The whole point of BV is to handle jumps that $W^{1,1}$ can't. A function like the characteristic function of a ball has a perfectly good "derivative" in the distributional sense — it's a surface measure on the boundary — but $\chi_E \notin W^{1,1}$ because that derivative isn't an L^1 function. BV says: fine, let the derivative be a measure. That's the entire motivation.

13.1 Definitions and first properties

Let $U \subseteq \mathbb{R}^n$ be open.

Definition 13.1 (Variation)**.** For $f \in L^1(U)$, define the *variation* of f in U by

$$\|Df\|(U) \;=\; \sup\left\{ \int_U f \,\operatorname{div}\varphi \, dx \;\middle|\; \varphi \in C_c^1(U;\mathbb{R}^n),\ |\varphi| \le 1 \right\}.$$

Say $f \in BV(U)$ if $\|Df\|(U) < \infty$.

So you test f against all smooth compactly supported divergence fields of magnitude at most 1, and take the sup. If that's finite, you're in BV. The sup is the total variation of Df, which by the Riesz representation theorem is an $\mathbb{R}^n$-valued Radon measure.

The norm on $BV(U)$ is $\|f\|_{BV} = \|f\|_{L^1} + \|Df\|(U)$.

Remark 13.2. $W^{1,1}(U) \subset BV(U)$. If $f \in W^{1,1}$, then $Df = \nabla f \, dx$ with $\nabla f \in L^1$, and $\|Df\|(U) = \int_U |\nabla f| \, dx < \infty$. But BV is strictly bigger: χ_E for E with smooth boundary is in BV but not $W^{1,1}$, because its derivative is a surface measure on ∂E and not an L^1 function.

Example 13.3. Some examples worth keeping in your head:

1. $f = \chi_E$ where E is a ball in $\mathbb{R}^n$. Then $f \in BV(\mathbb{R}^n)$ and $\|Df\|(\mathbb{R}^n) = \mathcal{H}^{n-1}(\partial E)$. The derivative is $(-\nu_E)\mathcal{H}^{n-1}\lfloor\partial E$ where ν_E is the outward unit normal. Clean.

2. The Cantor function on $[0,1]$: in BV but not $W^{1,1}$. Its derivative is entirely singular, concentrated on the Cantor set. No jump part either, it's all Cantor part.

3. Any monotone function $f : \mathbb{R} \to \mathbb{R}$ is BV_{loc}. This connects to the classical theory.

13.2 The structure theorem

OK so here's the structural result that makes everything else in the chapter go.

Theorem 13.4 (Structure Theorem for BV)**.** *Let $f \in BV(U)$. Then Df is an $\mathbb{R}^n$-valued Radon measure on U, admitting the decomposition*

$$Df \;=\; D^a f + D^s f,$$

where:

- *$D^a f$ is absolutely continuous with respect to $\mathcal{L}^n$. By Radon–Nikodym, $D^a f = \nabla f \, \mathcal{L}^n$ for some $\nabla f \in L^1(U; \mathbb{R}^n)$.*
- *$D^s f$ is singular with respect to $\mathcal{L}^n$: concentrated on a set of Lebesgue measure zero.*

This is basically just Lebesgue–Radon–Nikodym decomposition applied to the measure Df. Nothing deep yet, but naming the pieces is what lets you talk precisely about what's going on.

For χ_E, the absolutely continuous part vanishes and the singular part is the surface measure on ∂E. For a smooth function, it's all absolutely continuous and no singular part. For the devil's staircase, it's all singular.

You can further decompose $D^s f = D^j f + D^c f$ into a *jump part* concentrated on a countably $(n-1)$-rectifiable set and a *Cantor part* that's singular but diffuse. I'll come back to this in the pointwise properties section.

13.3 Approximation and compactness

Two big results. First, lower semicontinuity of the variation.

Theorem 13.5 (Lower semicontinuity)**.** *If $f_j \to f$ in $L^1(U)$, then*

$$\|Df\|(U) \;\le\; \liminf_{j\to\infty} \|Df_j\|(U).$$

Why? Because $\|Df\|(U)$ is defined as a supremum of continuous linear functionals of f. Each functional passes to the limit, and the sup of limits $\le$ the liminf of sups. Standard argument.

Theorem 13.6 (Approximation by smooth functions)**.** *If $f \in BV(U)$, there exist $f_j \in C^\infty(U) \cap BV(U)$ with*

$$f_j \to f \text{ in } L^1(U) \quad \text{and} \quad \|Df_j\|(U) \to \|Df\|(U).$$

Here's the deal: this is convergence in the *strict* topology, meaning L^1 convergence of functions plus convergence of total variation. You can NOT get $f_j \to f$ in the BV norm. That would require $\|Df_j - Df\|(U) \to 0$, which is impossible in general because BV functions can have singular parts and smooth functions can't. The mollification $f * \rho_\varepsilon$ converges in L^1 and the variation converges, but $D(f * \rho_\varepsilon) - Df$ doesn't go to zero as a measure.

BV is not the closure of C^∞ in the BV norm. That closure is $W^{1,1}$.

Theorem 13.7 (Compactness)**.** *Let $\{f_j\}$ be a sequence in $BV(U)$ with $\sup_j \|f_j\|_{BV(U)} < \infty$, where U is bounded with Lipschitz boundary. Then a subsequence converges in $L^1(U)$ to some $f \in BV(U)$.*

In Sobolev spaces you'd get weak convergence by reflexivity, but $W^{1,1}$ isn't reflexive and neither is BV. The compactness is only in L^1. Still, combined with lower semicontinuity, it's enough for the direct method. I hate this notation but it works.

13.4 Traces and extensions

Theorem 13.8 (Trace for BV). *Let $U \subset \mathbb{R}^n$ be bounded open with Lipschitz boundary. There's a bounded linear operator*

$$T : BV(U) \to L^1(\partial U; \mathcal{H}^{n-1})$$

with $Tf = f|_{\partial U}$ for $f \in BV(U) \cap C(\overline{U})$.

The trace only lands in $L^1(\partial U)$. No fractional Sobolev regularity. Makes sense: BV functions can jump, so the trace just records boundary values of those jumps. The proof goes through the same flattening machinery as Sobolev traces but needs the BV-specific approximation results.

Theorem 13.9 (Extension). *Let $U \subset \mathbb{R}^n$ be bounded with Lipschitz boundary. There's a bounded linear operator $E : BV(U) \to BV(\mathbb{R}^n)$ with $Ef = f$ a.e. in U, $\operatorname{spt}(Ef)$ compact, and $\|Ef\|_{BV(\mathbb{R}^n)} \leq C\|f\|_{BV(U)}$.*

Standard extension result. Mirrors the Sobolev case. The Lipschitz boundary condition is doing real work here.

14 October 23, 2024

Today: coarea formula and isoperimetric inequality. I think the coarea formula is the single best result in the chapter. Brian said the same thing last week.

14.1 Coarea formula for BV

Theorem 14.1 (Coarea formula). *Let $f \in BV(U)$. For $\mathcal{L}^1$-a.e. $t \in \mathbb{R}$, the level set $\{f > t\}$ has finite perimeter in U. And for every Borel set $A \subseteq U$,*

$$\|Df\|(A) = \int_{-\infty}^{\infty} \|\partial\{f > t\}\|(A)\,dt,$$

where $\|\partial\{f > t\}\|$ is the perimeter measure of the superlevel set $\{x \in U : f(x) > t\}$.

Let me unpack this. The left side is the total variation of Df on A, a single number measuring how much f oscillates. The right side slices this into contributions from each level: at height t, the level set $\{f > t\}$ contributes its perimeter restricted to A. Sum over all heights and you recover the total variation.

For $f \in W^{1,1}(U)$, the formula becomes

$$\int_A |\nabla f|\,dx = \int_{-\infty}^{\infty} \mathcal{H}^{n-1}(A \cap \partial^*\{f > t\})\,dt,$$

which is a generalization of the classical coarea formula from differential geometry.

Remark 14.2. The formula also gives a converse criterion: $f \in L^1(U)$ is in $BV(U)$ if and only if $\int_{-\infty}^{\infty} \|\partial\{f > t\}\|(U)\,dt < \infty$. Sometimes this is the easiest way to check BV membership.

I found the proof confusing on first read because Evans uses the approximation theorem to reduce to smooth f first, then proves the smooth case by the classical coarea formula, then passes to the limit using lower semicontinuity. The inequality $\leq$ and $\geq$ are proved separately, and the $\geq$ direction is the harder one.

14.2 Isoperimetric inequalities

Among all sets of given volume, the ball has the smallest perimeter. Known since antiquity. Here's the precise statement.

Theorem 14.3 (Isoperimetric inequality)**.** *Let $E \subset \mathbb{R}^n$ have finite perimeter. Then*

$$\|\partial E\|(\mathbb{R}^n) \;\geq\; n\,\alpha(n)^{1/n}\left(\mathcal{L}^n(E)\right)^{(n-1)/n},$$

where $\alpha(n) = \mathcal{L}^n(B(0,1))$. Equality iff E is a ball (up to measure zero).

The argument uses the coarea formula. You first establish the BV Sobolev inequality: for $f \in BV(\mathbb{R}^n)$,

$$\|f\|_{L^{n/(n-1)}(\mathbb{R}^n)} \;\leq\; \frac{1}{n\,\alpha(n)^{1/n}}\,\|Df\|(\mathbb{R}^n).$$

Then plug in $f = \chi_E$. The left side gives $\mathcal{L}^n(E)^{(n-1)/n}$ and the right side gives $\|\partial E\|(\mathbb{R}^n)$ times the constant. Done.

Well, not quite done. The equality case requires a symmetrization argument, which is a different beast entirely. But the inequality itself follows from the coarea formula in about half a page.

There's also a *relative isoperimetric inequality*: if U is bounded Lipschitz and $\mathcal{L}^n(E \cap U) \leq \frac{1}{2}\mathcal{L}^n(U)$, then

$$\mathcal{L}^n(E \cap U)^{(n-1)/n} \;\leq\; C(U)\;\|\partial E\|(U).$$

Useful for compactness arguments.

15 October 28, 2024

This chapter is harder than Chapter 4. Today we get into the geometric measure theory proper: reduced boundary, blow-up, and the generalized divergence theorem.

15.1 Sets of finite perimeter and the reduced boundary

The idea: for a set E of finite perimeter, the "boundary" in the BV sense isn't the topological boundary ∂E (which can be wild). It's a well-behaved subset called the *reduced boundary $\partial^* E$.*

Definition 15.1 (Reduced boundary)**.** Let E have finite perimeter in $\mathbb{R}^n$. Write $\mu_E = D\chi_E$ for the Gauss–Green measure and $\|\mu_E\| = \|\partial E\|$ for its total variation. A point $x \in \mathbb{R}^n$ belongs to the *reduced boundary $\partial^* E$* if:

1. $\|\mu_E\|(B(x,r)) > 0$ for all $r > 0$.
2. The limit $\nu_E(x) = \lim_{r\to 0} \frac{\mu_E(B(x,r))}{\|\mu_E\|(B(x,r))}$ exists.
3. $|\nu_E(x)| = 1$.

The vector $\nu_E(x)$ is called the *measure-theoretic outer unit normal.*

So $\partial^* E$ consists of points where the boundary "looks like a hyperplane" at small scales. At every such point there's a meaningful notion of which way is out. The reduced boundary $\partial^* E$ is always contained in the topological boundary ∂E, and they can differ a lot. If you modify E on a null set, the topological boundary can change wildly but $\partial^* E$ stays the same.

15.2 The blow-up theorem

This is the hard direction. The proof is genuinely hard and I don't have a clean two-line summary.

Theorem 15.2 (Blow-up at the reduced boundary). *Let E have finite perimeter, $x \in \partial^* E$ with outer normal $\nu_E(x)$. Define*

$$E_{x,r} \;=\; \frac{E-x}{r}.$$

Then $\chi_{E_{x,r}} \to \chi_{H^-(\nu_E(x))}$ in $L^1_{\text{loc}}(\mathbb{R}^n)$ as $r \to 0^+$, where $H^-(\nu) = \{y : y \cdot \nu < 0\}$.

You zoom in at a point of $\partial^* E$ and the set converges to a half-space. The boundary looks flat at infinitesimal scales, and the normal to the half-space is exactly $\nu_E(x)$ from the definition. The proof uses Besicovitch differentiation, the definition of the reduced boundary, and BV compactness. Evans pp. 190–200.

Remark 15.3. A major consequence: $\partial^* E$ is countably $(n-1)$-rectifiable, and the perimeter measure satisfies $\|\partial E\| = \mathcal{H}^{n-1}\lfloor\partial^* E$. The perimeter measure is $(n-1)$-dimensional Hausdorff measure restricted to the reduced boundary. This is deep.

15.3 Measure-theoretic boundary and Gauss–Green

Definition 15.4 (Measure-theoretic boundary). The *measure-theoretic boundary* $\partial_* E$ consists of all $x \in \mathbb{R}^n$ where neither E nor its complement has density 1:

$$\partial_* E \;=\; \left\{ x : \limsup_{r\to 0} \frac{\mathcal{L}^n(E \cap B(x,r))}{\alpha(n) r^n} > 0 \text{ and } \limsup_{r\to 0} \frac{\mathcal{L}^n(B(x,r) \setminus E)}{\alpha(n) r^n} > 0 \right\}.$$

We always have $\partial^* E \subseteq \partial_* E \subseteq \partial E$. For sets of finite perimeter, $\mathcal{H}^{n-1}(\partial_* E \setminus \partial^* E) = 0$, so the reduced and measure-theoretic boundaries agree up to $\mathcal{H}^{n-1}$-null sets.

Now the payoff.

Theorem 15.5 (Generalized Gauss–Green theorem). *Let $E \subset \mathbb{R}^n$ have finite perimeter, $\varphi \in C^1_c(\mathbb{R}^n; \mathbb{R}^n)$. Then*

$$\int_E \operatorname{div} \varphi \, dx \;=\; \int_{\partial^* E} \varphi \cdot \nu_E \, d\mathcal{H}^{n-1}.$$

The classical divergence theorem needs E to have smooth or Lipschitz boundary. This version only needs finite perimeter. The integral over the boundary is over $\partial^* E$ with the measure-theoretic normal. I think the cleanest way to see why this works is: $D\chi_E = -\nu_E \mathcal{H}^{n-1}\lfloor\partial^* E$ by the structure theory, and then the formula is just the definition of distributional derivative applied to χ_E.

15.4 Pointwise properties of BV functions

What can you say about individual points? More than you'd expect.

Theorem 15.6 (Approximate limits and jump set). *Let $f \in BV(U)$. Then:*

1. *For $\mathcal{H}^{n-1}$-a.e. $x \in U$, f has an approximate limit $\tilde{f}(x) = \operatorname{ap}\lim_{y\to x} f(y)$ (the precise representative).*
2. *The jump set J_f is countably $(n-1)$-rectifiable.*
3. *$D^s f = D^j f + D^c f$, where $D^j f = (f^+ - f^-)\nu_f \, \mathcal{H}^{n-1}\lfloor J_f$ is the jump part and $D^c f$ is the Cantor part.*

So the full decomposition is:

$$Df \;=\; \underbrace{\nabla f\,\mathcal{L}^n}_{\text{abs. cont.}} \;+\; \underbrace{(f^+ - f^-)\nu_f\,\mathcal{H}^{n-1}\lfloor J_f}_{\text{jump}} \;+\; \underbrace{D^c f}_{\text{Cantor}}\;.$$

The three parts live on "different" sets. The ac part is spread over U. The jump part sits on the $(n-1)$-dimensional jump set. And the Cantor part is on something even weirder, like the Cantor set itself.

For χ_E with E having finite perimeter: $\nabla\chi_E = 0$, $D^c\chi_E = 0$, and $D^j\chi_E = -\nu_E\,\mathcal{H}^{n-1}\lfloor\partial^* E$. It's all jump, nothing fancy.

15.5 Essential variation on lines

Theorem 15.7 (Characterization via sections)**.** *$f \in L^1(U)$ is in $BV(U)$ if and only if for each coordinate direction e_i, the one-dimensional slices $t \mapsto f(x' + te_i)$ have finite essential variation for $\mathcal{L}^{n-1}$-a.e. x', and the total of these variations is finite.*

This is the BV analogue of the ACL characterization for Sobolev functions. For Sobolev, the slices are absolutely continuous; for BV, the slices have bounded essential variation.

15.6 Criterion for finite perimeter

Theorem 15.8 (Criterion)**.** *$E \subset \mathbb{R}^n$ measurable has finite perimeter in U iff*

$$\sup\left\{\int_E \operatorname{div}\varphi\,dx \;\middle|\; \varphi \in C^1_c(U;\mathbb{R}^n),\ |\varphi| \le 1\right\} < \infty,$$

which is the same as $\chi_E \in BV(U)$.

Sets of finite perimeter include: sets with Lipschitz boundary, convex sets, sublevel sets $\{f < t\}$ of BV functions for a.e. t (by the coarea formula). Sets that do NOT have finite perimeter: fractal sets whose boundary has infinite $\mathcal{H}^{n-1}$-measure, like the Koch snowflake in $\mathbb{R}^2$.

16 November 4, 2024

I'm tired. Four chapters of measure theory will do that to you. But this chapter is where it all pays off: Sobolev functions, BV functions, convex functions — all the rough objects from earlier — turn out to be differentiable in various precise senses almost everywhere. And then the punchline: Sobolev functions are "almost C^1."

16.1 L^p differentiability

The idea is simple. Instead of asking that f be differentiable pointwise, you ask that the Taylor remainder is small *on average* over shrinking balls.

Definition 16.1 (L^p differentiability)**.** Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $1 \le p \le \infty$. We say f is L^p *differentiable* at x if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}$ such that

$$\left(\frac{1}{r^n}\int_{B(x,r)} |f(y) - f(x) - L\cdot(y-x)|^p \,\mathrm{d}y\right)^{1/p} = o(r) \quad \text{as } r \to 0.$$

When this holds, L is the L^p derivative of f at x.

Note the scaling: the left side has units of f, and $o(r)$ means the remainder decays faster than linear. Classical differentiability is the L^∞ case, so L^p differentiability is weaker for finite p.

Theorem 16.2 (L^{p^*} differentiability of $W^{1,p}$ functions)**.** *If $f \in W^{1,p}_{\mathrm{loc}}(\mathbb{R}^n)$ with $1 \le p < n$, then f is L^{p^*} differentiable at $\mathcal{L}^n$-a.e. x, where $p^* = np/(n-p)$. The L^{p^*} derivative equals $Df(x)$.*

The proof uses Morrey's estimate from Chapter 4. You write f as its average plus an integral of the gradient, hit it with Poincaré on balls, and Lebesgue differentiation pins down the base point. The exponent p^* is sharp. You don't get L^q for $q > p^*$.

16.2 Approximate differentiability

OK so this is a weird object at first. The approximate derivative says: f looks differentiable if you ignore a set of density zero.

Definition 16.3 (Approximate differentiability)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$.

1. The *approximate limit* $\ell = \operatorname{ap}\lim_{y\to x} f(y)$ exists if for every $\varepsilon > 0$,

$$\lim_{r\to 0} \frac{\mathcal{L}^n(\{y \in B(x,r) : |f(y) - \ell| \ge \varepsilon\})}{\alpha(n) r^n} = 0.$$

   The bad set has density zero at x.

2. f is *approximately differentiable* at x if there's a linear $L : \mathbb{R}^n \to \mathbb{R}$ with

$$\operatorname{ap}\lim_{y\to x} \frac{f(y) - f(x) - L \cdot (y-x)}{|y-x|} = 0.$$

   Write $\operatorname{ap} Df(x) = L$.

You're replacing the ordinary limit with an approximate limit. The set where the difference quotient is large could be nonempty, even uncountable. You just need it to be thin in a density sense.

Theorem 16.4. *If f is approximately differentiable at x, then* $\operatorname{ap} Df(x)$ *is unique.*

Proof. Suppose L_1, L_2 both work. For each $\varepsilon > 0$, the sets $A_i = \{y : |f(y) - f(x) - L_i(y-x)|/|y-x| \ge \varepsilon\}$ each have density zero at x. Their union also has density zero. On $B(x,r) \setminus (A_1 \cup A_2)$ we get $|(L_1 - L_2)(y-x)| \le 2\varepsilon|y-x|$, and this set has density one. Pick directions and send $\varepsilon \to 0$. □

Now the big result for BV.

Theorem 16.5 (BV functions are approximately differentiable a.e.)**.** *Let $f \in BV_{\mathrm{loc}}(\mathbb{R}^n)$. Then f is approximately differentiable at $\mathcal{L}^n$-a.e. x, with*

$$\operatorname{ap} Df(x) = \nabla f(x) \quad \mathcal{L}^n\text{-a.e.},$$

where ∇f is the Radon–Nikodým derivative of Df with respect to $\mathcal{L}^n$.

This is surprisingly strong. BV functions can jump across hypersurfaces, but away from those jumps they're approximately differentiable with the "expected" gradient. The jumps form a set of $\mathcal{L}^n$-measure zero and hence density zero. The approximate derivative just ignores them.

Remark 16.6. The hierarchy, strongest to weakest:

$$\text{classically diff.} \implies L^\infty \text{ diff.} \implies L^p \text{ diff.} \implies \text{approx. diff.}$$

None of the arrows reverse.

17 November 6, 2024

Short lecture today. We do the $p > n$ case, Rademacher redux, and convex functions.

17.1 Differentiability a.e. for $W^{1,p}$, $p > n$

When $p > n$ you jump all the way to classical differentiability. This is where the Sobolev embedding really pays off.

Theorem 17.1. *Let $f \in W^{1,p}_{\mathrm{loc}}(\mathbb{R}^n)$ with $n < p \leq \infty$. Then f is differentiable at $\mathcal{L}^n$-a.e. x, with derivative $Df(x)$.*

Proof. By Morrey's inequality, for $y \in B(x,r)$:

$$|f(y) - f(x) - Df(x) \cdot (y-x)| \leq Cr \left(\frac{1}{r^n} \int_{B(x,r)} |Df(y) - Df(x)|^p \, \mathrm{d}y \right)^{1/p}.$$

Lebesgue differentiation applied to $|Df - Df(x)|^p$ gives the right side as $o(r)$ for a.e. x. That's classical differentiability. □

Short and clean. Morrey does the heavy lifting.

Theorem 17.2 (Rademacher's Theorem — second proof)**.** *Every locally Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable $\mathcal{L}^n$-a.e.*

Proof. Locally Lipschitz implies $f \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{R}^n)$. Since $\infty > n$, apply Theorem 17.1. □

Two lines. The first proof in Chapter 3 was this painful argument with directional derivatives and Fubini. This proof hides the pain in the Chapter 4 prerequisites, but whatever, it works.

17.2 Convex functions

Theorem 17.3. *Every convex function $f : \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz.*

Convexity is a global geometric condition but it forces local analytic regularity. A convex function can't oscillate too wildly. Combined with Rademacher: every convex function is differentiable a.e. But Aleksandrov goes one derivative further.

17.3 Aleksandrov's theorem

This result is amazing. A convex function has second derivatives almost everywhere.

Theorem 17.4 (Aleksandrov)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex. For $\mathcal{L}^n$-a.e. x, there exists a symmetric $D^2 f(x) \in \mathbb{R}^{n \times n}$ with*

$$\lim_{r \to 0} \frac{1}{r^n} \int_{B(x,r)} \frac{\left| f(y) - f(x) - Df(x) \cdot (y-x) - \frac{1}{2}(y-x)^T D^2 f(x)(y-x) \right|}{r^2} \, \mathrm{d}y = 0.$$

And $D^2 f(x) \geq 0$ a.e.

I mean, think about what this says. The second-order Taylor remainder, averaged over a ball, is $o(r^2)$. Convex functions have a second-order Taylor expansion a.e., and the Hessian is nonneg. That's what you'd expect from convexity, but proving it exists without any smoothness assumption is the hard part.

Proof sketch. The argument takes all of Evans Section 6.4. Three steps:

1. f is locally Lipschitz by Theorem 17.3, so differentiable a.e. by Rademacher. The gradient Df is BV (the distributional Hessian is a nonneg. matrix-valued Radon measure).
2. Decompose $D^2 f$ into ac and singular parts using BV structure.
3. The Taylor expansion holds at Lebesgue points of the ac part, which is a.e.

The jump from first to second derivatives is the deepest result in the chapter. □

Look, $f(x) = |x|$ on $\mathbb{R}$ has $f''(x) = 0$ for $x \neq 0$ and isn't twice differentiable at $x = 0$. But that's a single point, measure zero. Aleksandrov says this is always the situation for convex functions.

17.4 Whitney's extension theorem

Theorem 17.5 (Whitney extension). *Let $C \subseteq \mathbb{R}^n$ be closed. Suppose $f : C \to \mathbb{R}$ and $d : C \to \mathbb{R}^n$ satisfy the compatibility condition: for every $\varepsilon > 0$ and compact $K \subseteq \mathbb{R}^n$, there exists $\delta > 0$ such that*

$$|f(y) - f(x) - d(x) \cdot (y - x)| \leq \varepsilon\, |y - x|$$

for $x, y \in C \cap K$ with $|x - y| < \delta$.

Then there exists $F \in C^1(\mathbb{R}^n)$ with $F|_C = f$ and $DF|_C = d$.

I won't go through the proof. It's a partition-of-unity construction with Whitney cubes. The point is the statement: if your data (f, d) on C is *consistent* with being a function and its derivative, then an actual C^1 extension exists.

17.5 Approximation by C^1 functions

Here's the punchline of the entire book.

Theorem 17.6 (C^1 approximation). *Let $f \in W^{1,p}(\mathbb{R}^n)$, $1 \leq p \leq \infty$. For every $\varepsilon > 0$, there's a $g \in C^1(\mathbb{R}^n)$ with*

$$\mathcal{L}^n(\{x : f(x) \neq g(x) \text{ or } Df(x) \neq Dg(x)\}) < \varepsilon.$$

Read that again. It's not saying f is close to g in some norm. It's saying f literally *equals* a C^1 function except on a set of measure less than ε. The function and its gradient agree with a smooth function outside a tiny exceptional set.

Proof sketch. Three ingredients, nothing more:

1. **Approximate differentiability a.e.** By Theorem 16.2 or Theorem 16.5, f is approximately differentiable a.e. with approximate derivative equal to Df.
2. **Lusin-type extraction.** Egorov's theorem plus the definition of approximate differentiability gives a closed set C with $\mathcal{L}^n(\mathbb{R}^n \backslash C) < \varepsilon$ such that $f|_C$ and $Df|_C$ satisfy Whitney's compatibility condition.
3. **Whitney extension.** Apply Theorem 17.5 to $(f|_C, Df|_C)$ to get $g \in C^1(\mathbb{R}^n)$ with $g = f$ and $Dg = Df$ on C.

The exceptional set is $\mathbb{R}^n \setminus C$, which has measure $< \varepsilon$. □

This is a Lusin-type theorem for Sobolev functions. Lusin says a measurable function is continuous outside a small set. This says a Sobolev function is C^1 outside a small set.

Remark 17.7. This is NOT mollification. Mollification gives $f_\varepsilon \in C^\infty$ with $f_\varepsilon \to f$ in $W^{1,p}$, but $f_\varepsilon \neq f$ everywhere. The C^1 approximation theorem gives pointwise equality on a big set. Much stronger, but you only get C^1 out, not C^∞.

Why does this matter? It gives an intrinsic characterization of Sobolev spaces: $W^{1,p}$ functions are exactly the L^p functions that agree with C^1 functions outside sets of small measure, with the right gradient control. It's used in geometric measure theory to transfer smooth results to Sobolev functions. And for PDEs, it sometimes lets you reduce regularity arguments to the C^1 case.

Everything in Chapter 6 depends on the hard analysis from Chapters 3–5. The Sobolev embeddings, the BV theory, the fine properties of measures. This is where those investments pay off, and the C^1 approximation theorem is the crown jewel.

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